

Building Instructions Mushroom Substrate Mixer and Portioning Device

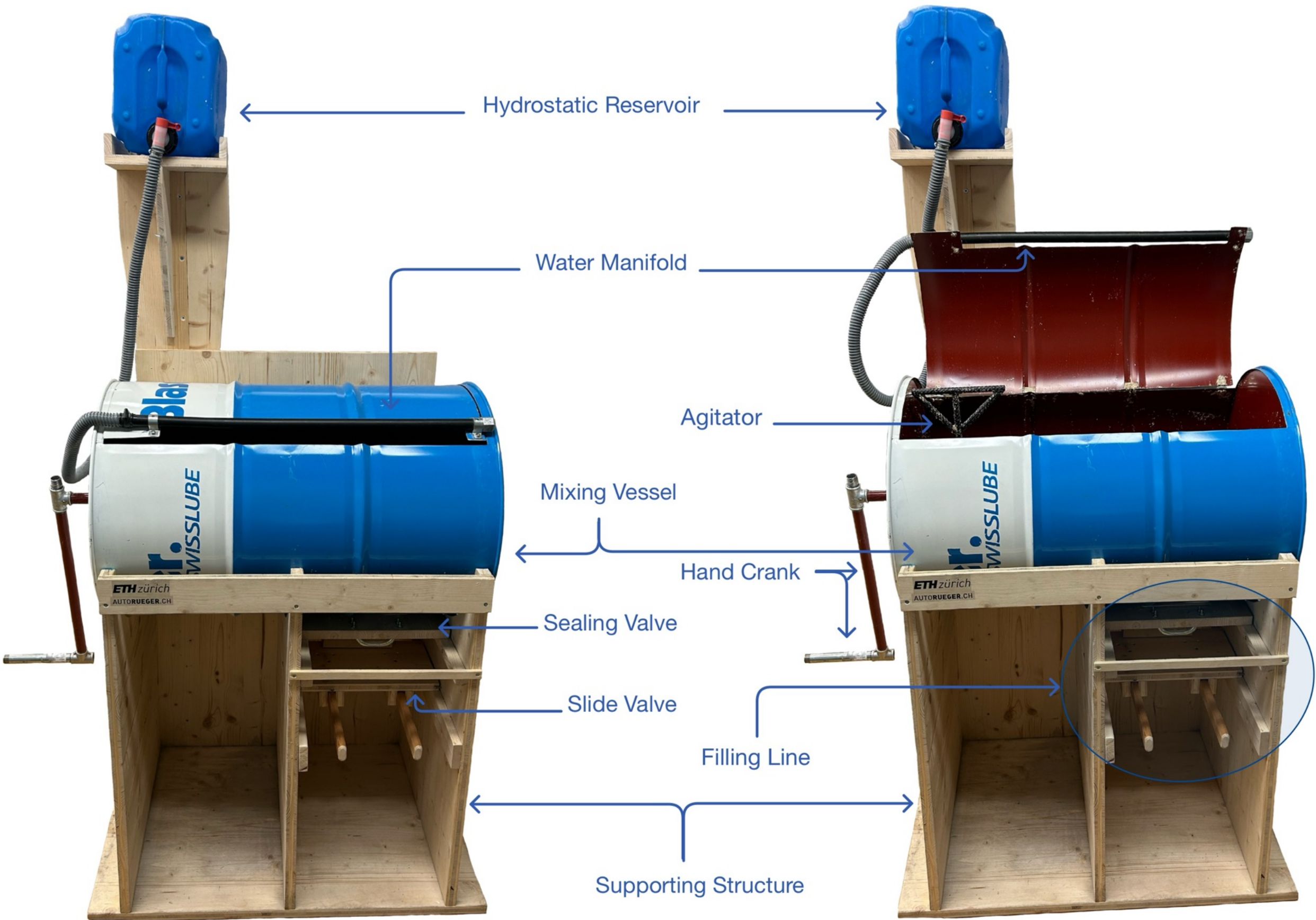
Based on the prototype developed during the bachelor's thesis

Product Development of a Substrate Mixing and Portioning Device
at ETH Zurich – Global Health Engineering

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User Manual



Full view of the Mushroom Substrate Mixer and Portioning Device.

Introduction

This machine helps you prepare mushroom substrate. You can mix dry materials such as sawdust, cotton husks, straw, coffee grounds, or wood chips, add water, and then fill the finished substrate into growing bags.

The machine is low-cost, hand-powered, easy to repair, and built from common materials. It does not require electricity or fuel. All building plans are included in this document, and you are welcome to rebuild or improve the machine. A video on how to use the device is available [here](#).

Be creative: This plan is merely a suggestion that performed well during testing at the Swiss laboratory. If certain materials or parts are not available to you, feel free to substitute them with others. The notes accompanying the technical drawings indicate which dimensions should not be changed and which can be adjusted.

Overview of the Machine

- **Mixing Barrel:** Inside the barrel is an agitator (mixing paddle). You turn it using the hand crank.
- **Water Tank and Manifold:** The blue tank (jerrycan) stores water. Water flows through a hose into a black pipe with holes (manifold), which spreads the water evenly across the substrate.
- **Portioning System:** This system helps you fill bags faster and with similar portion sizes. It uses sliding plates and a pyramid-shaped diffuser.

Preparing a Batch of Substrate

- Add your dry substrate ingredients into the mixer. Do not exceed the maximum fill level.
- Turn the hand crank slowly and steadily.
- After a few rotations, the dry materials will already be well mixed.

IMPORTANT: Do not fill the barrel above the height of the drive shaft. This is the maximum safe fill level. If the oil drum is overfilled, the substrate will clog the drum and it will no longer be possible to stir it.

Adding Water

- Place the water canister on top of the wooden stand.
- Open the valve so water flows into the mixer.
- Keep mixing the whole time while water enters.
- The water tank empties in about five minutes.

How much water to add: A good substrate has 60–70% water content. Water needed $\approx 1.3 \times$ weight of dry materials. Can vary depending on the substrate.

Checking Moisture — The Squeeze Test

Take a handful of substrate and squeeze it.

- A few drops of water flow out = perfect.
- Many drops flow out = too wet.
- No water flows out, substrate feels dry = add more water.

Filling the Growing Bags

Starting the filling process:

- Pull the upper plate (sealing valve) forward to open the filling line.
- Place a bag or bucket under the outlet.
- For cleaner filling, pull the lower plate (slide valve) together with the upper plate (sealing valve).

Filling the bag:

- Hold a plastic bag between the two wooden bars of the slide valve.
- Push the plate backward so substrate falls into the bag.
- Pull the plate forward again to close the outlet.

Portion size: One portion is about 1.5 kg. Fill the same bag several times for bigger portions.

Important Tip

Turn the agitator once between every bag.

Without it, the substrate won't flow into the bagging mechanism.

Cycle Times

- Mixing and watering: about 5 minutes
- Portioning: 5–10 minutes (two workers), or 15 minutes (one worker)

After Filling

- Pasteurize bags with steam
- Cool bags
- Inoculate bags with mushroom spawn
- Incubate mushroom spawn in bags
- Wait for mushrooms to harvest

Construction Manual

Introduction

This construction manual explains how the mushroom substrate mixer and portioning device is built. All building plans, photos of the final prototype, and the technical drawings are included in this document.

Overview of the Four Assemblies

- **1. Supporting Structure and Mixing Vessel**
Wooden frame + 200-liter (55-gal) steel drum.
- **2. Agitator (Mixing Paddle System)**
Steel pipes + threaded connectors + welded rebars.
- **3. Watering System**
20-liter jerrycan + wooden stand + water hose + perforated plastic pipe.
- **4. Portioning System**
Wooden sliding plates + pyramid diffuser.

Technical drawings of all parts and pictures follow on the next pages.

Required Skills

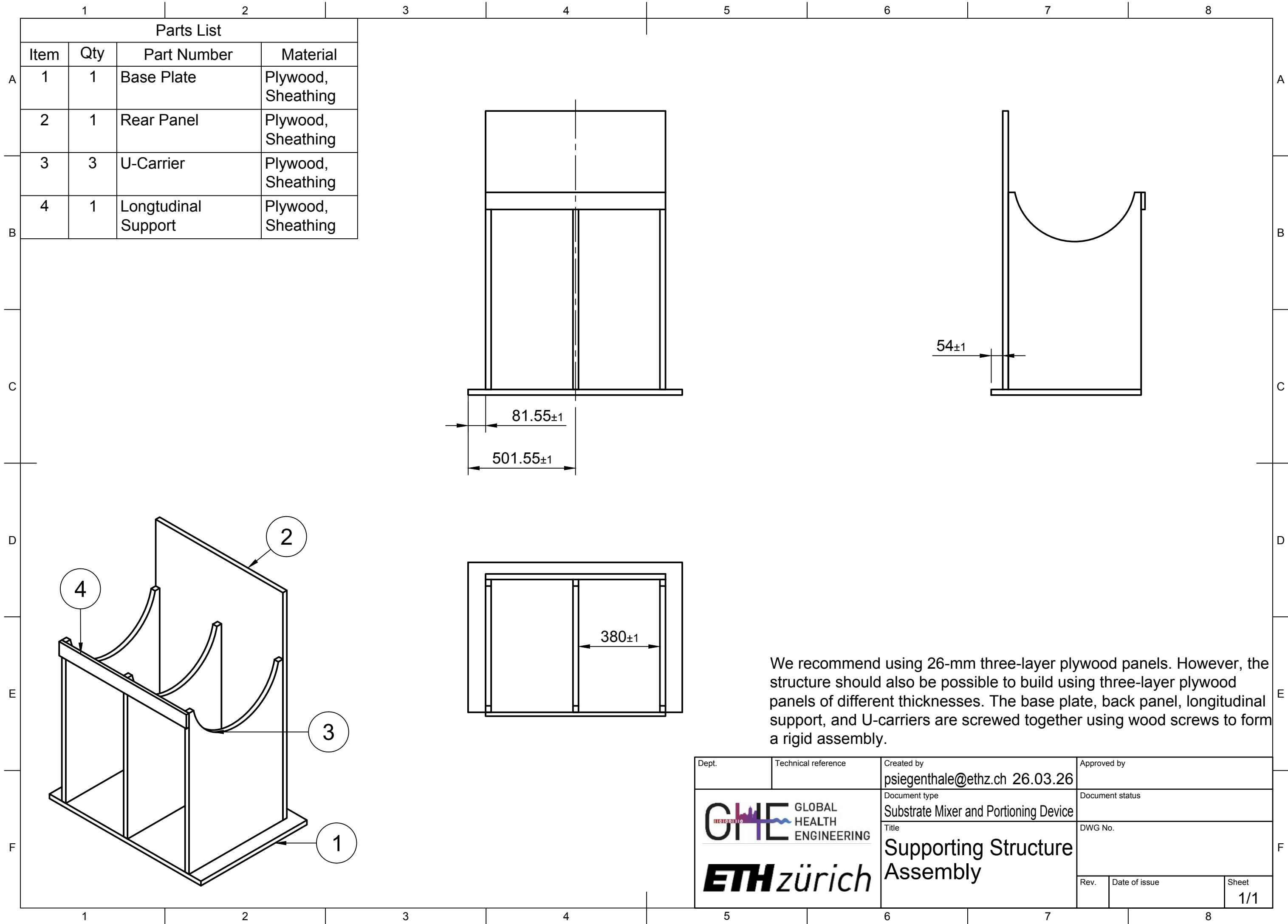
- Basic woodworking
- Basic welding
- Simple plumbing

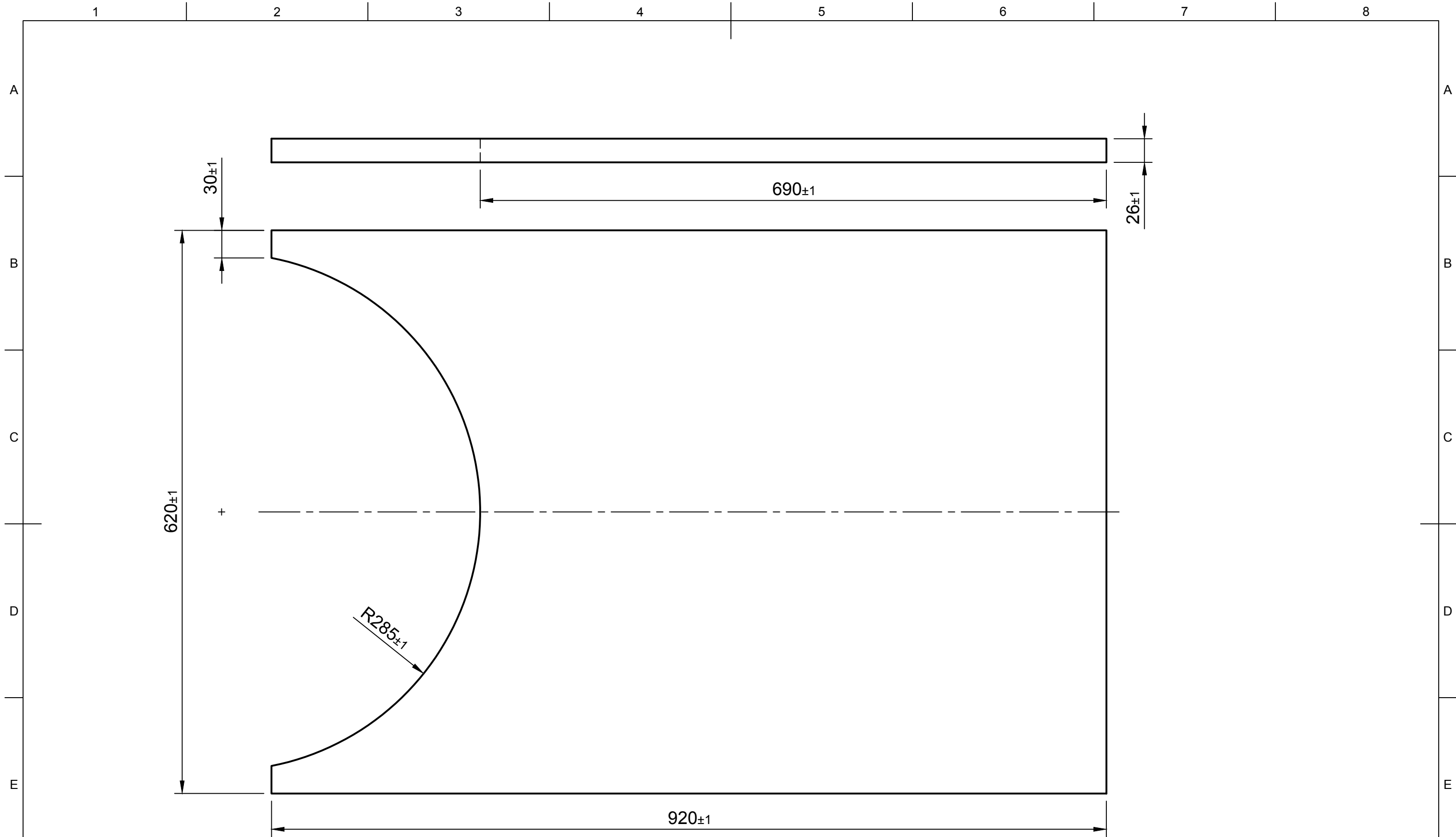
Don't Forget Rust and Moisture Protection

- All wooden components must be painted (lacquered) to protect them from moisture.
- All steel and sheet metal parts must be sprayed or lacquered with rust inhibitor.

Assembly Instructions – Technical Drawings

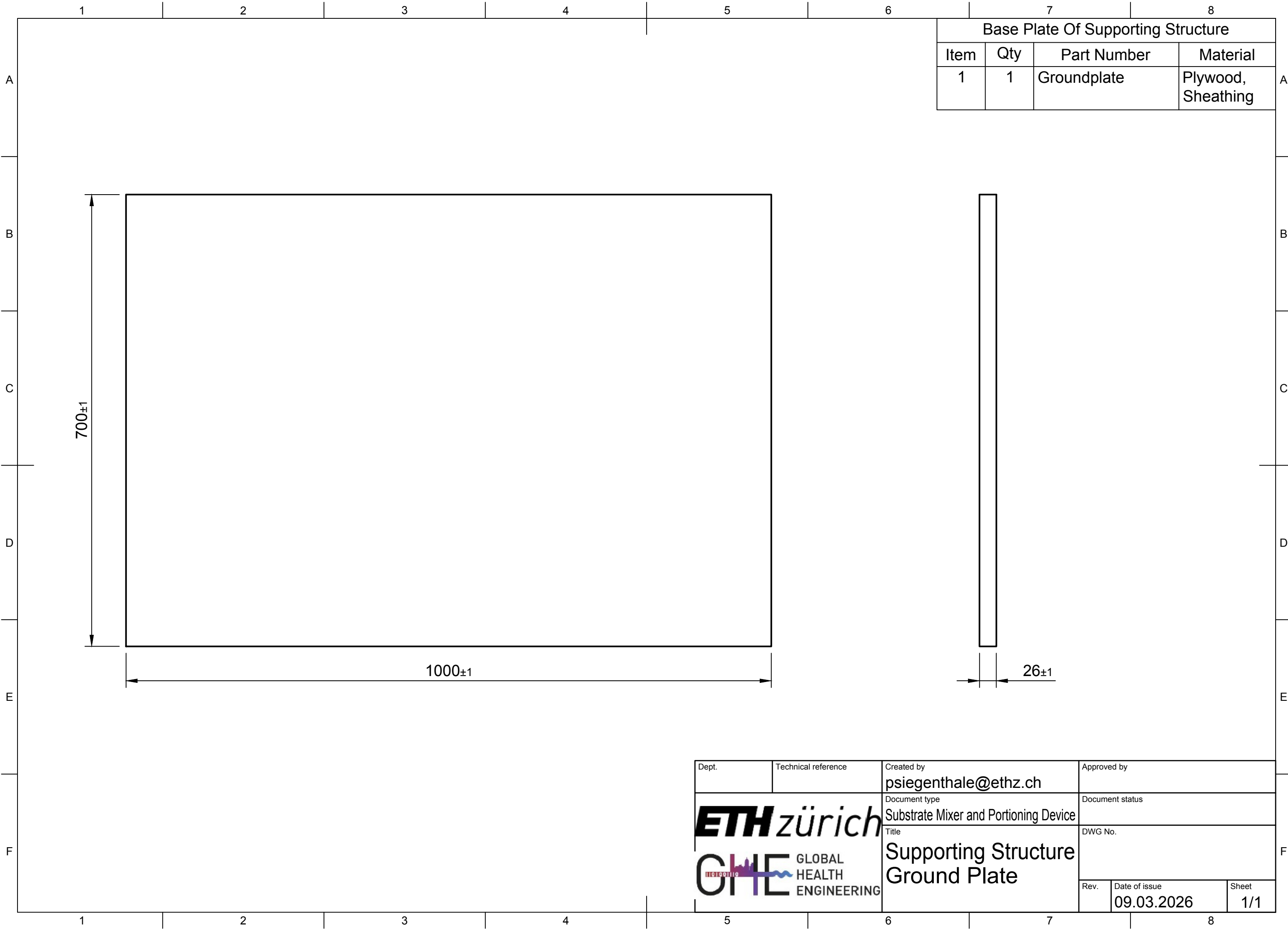
Supporting Structure

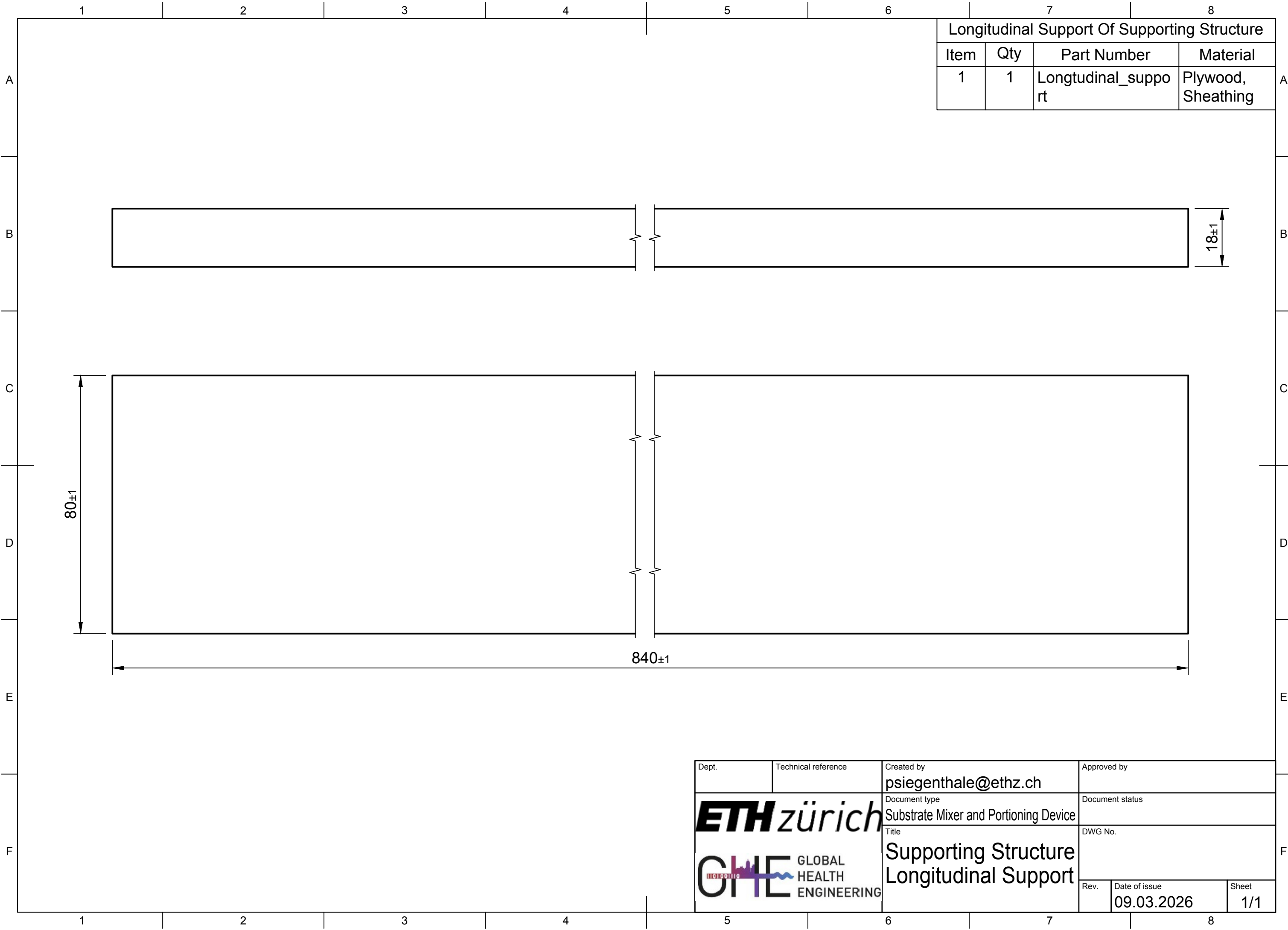




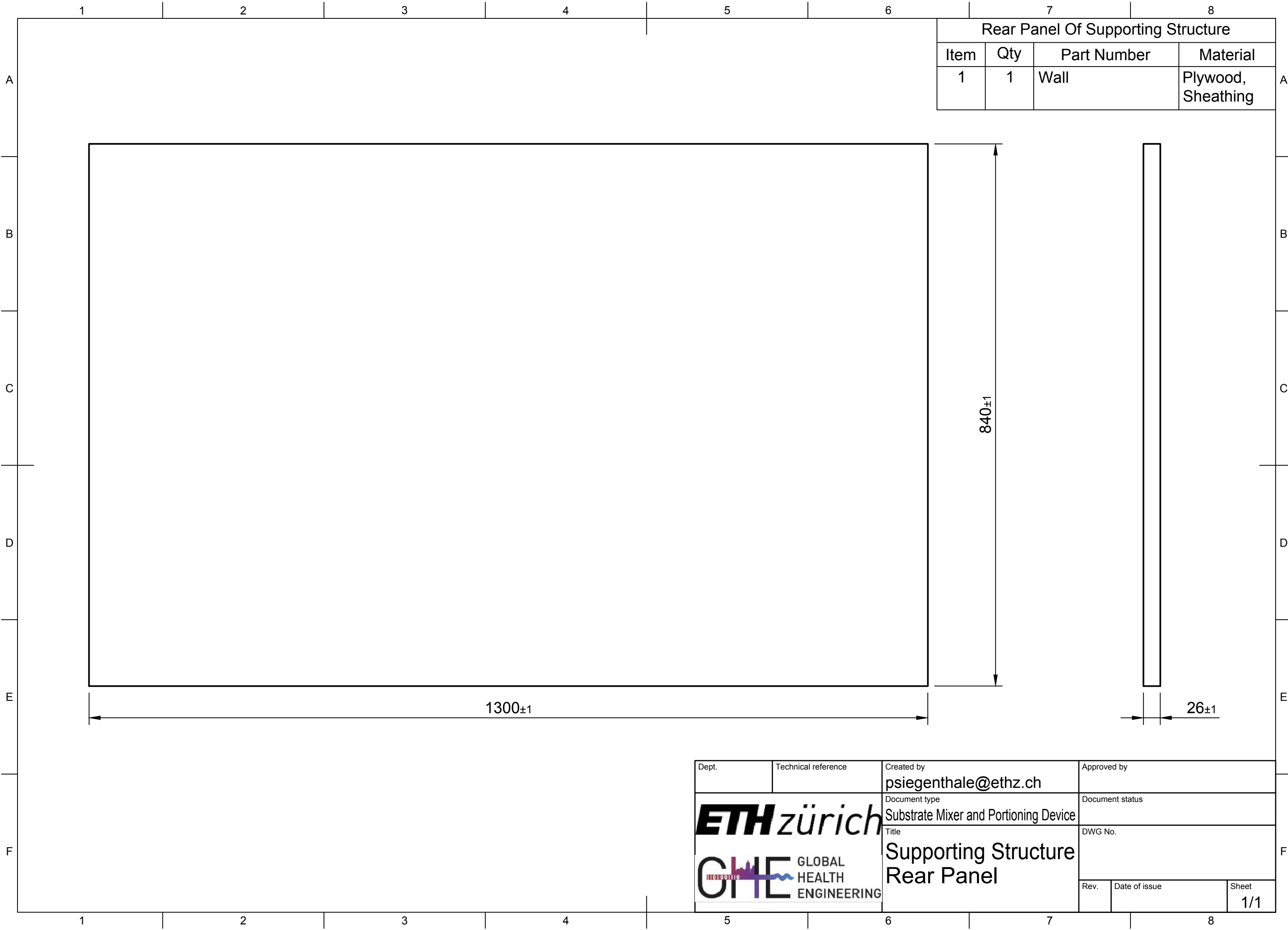
U-Carrier Of Supporting Structure		
Item	Part Number	Material
1	U-carrier	Plywood, Sheathing

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 		Document type Substrate Mixer and Portioning Device	Document status	
		Title Supporting Structure U-Carrier	DWG No.	
			Rev.	Date of issue 09.03.2026
			Sheet	1/1

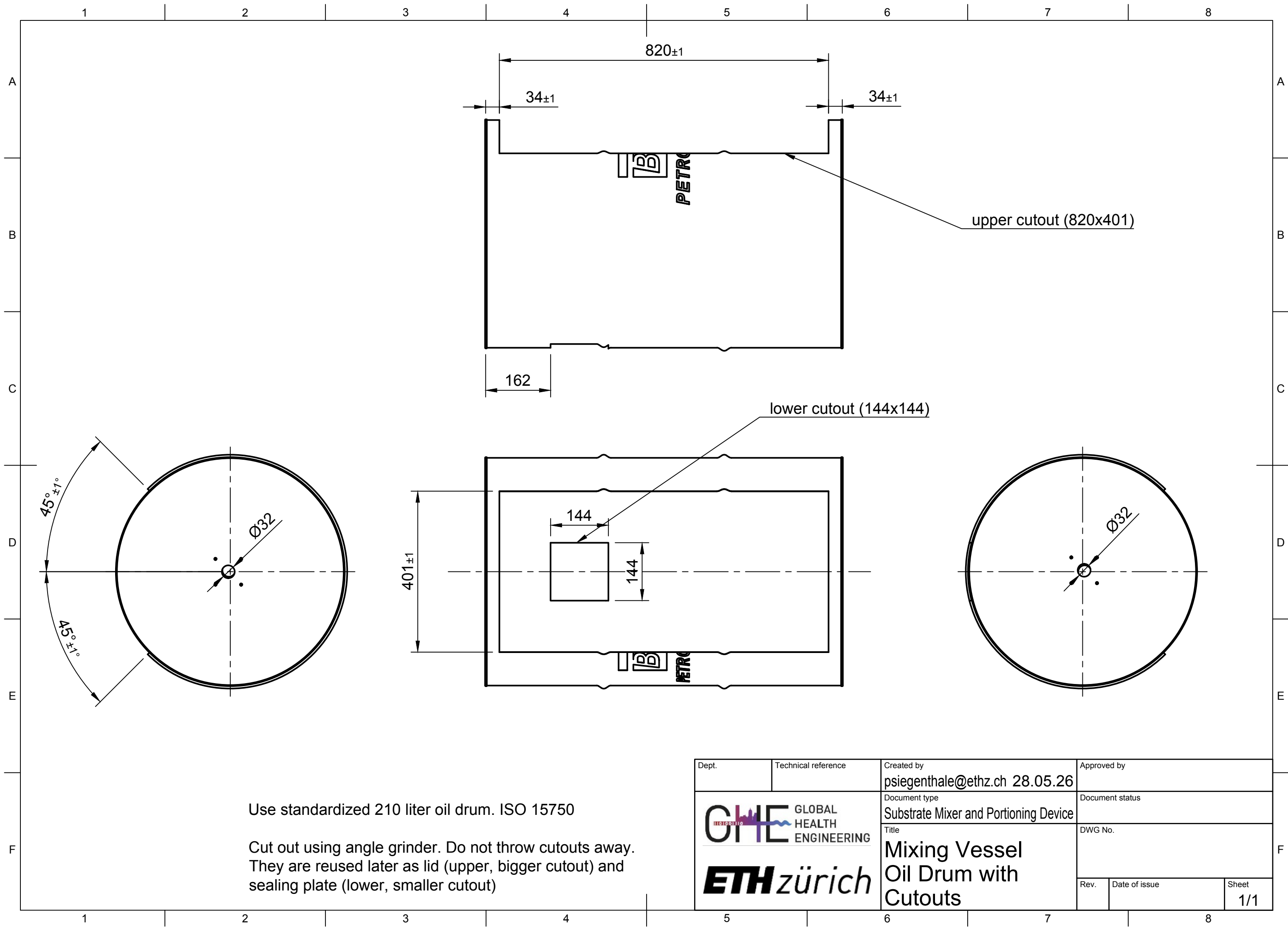




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ETHzürich GHIE GLOBAL HEALTH ENGINEERING	Document type Substrate Mixer and Portioning Device		Document status
	Title Supporting Structure Longitudinal Support		DWG No.
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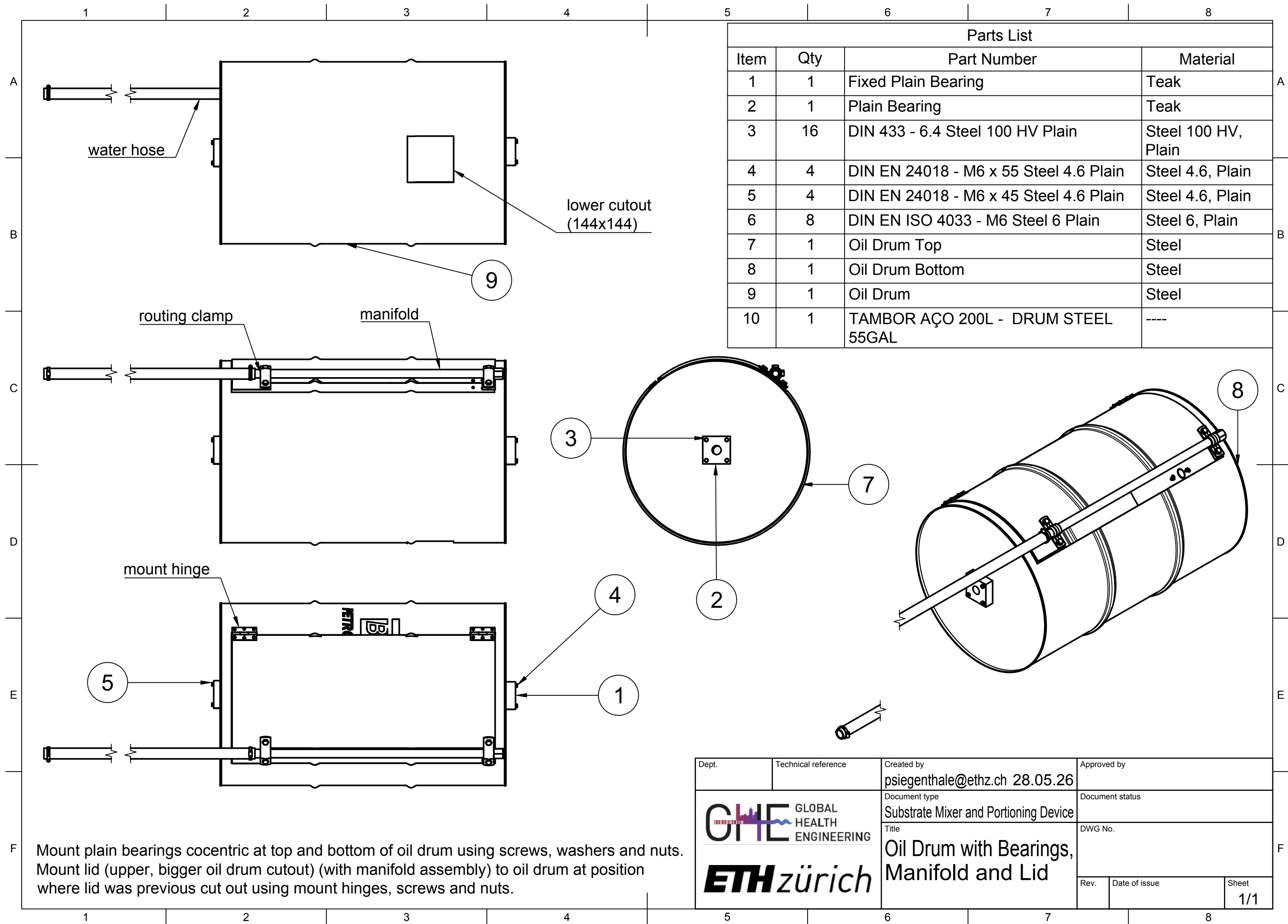
Oil Drum with Cutouts

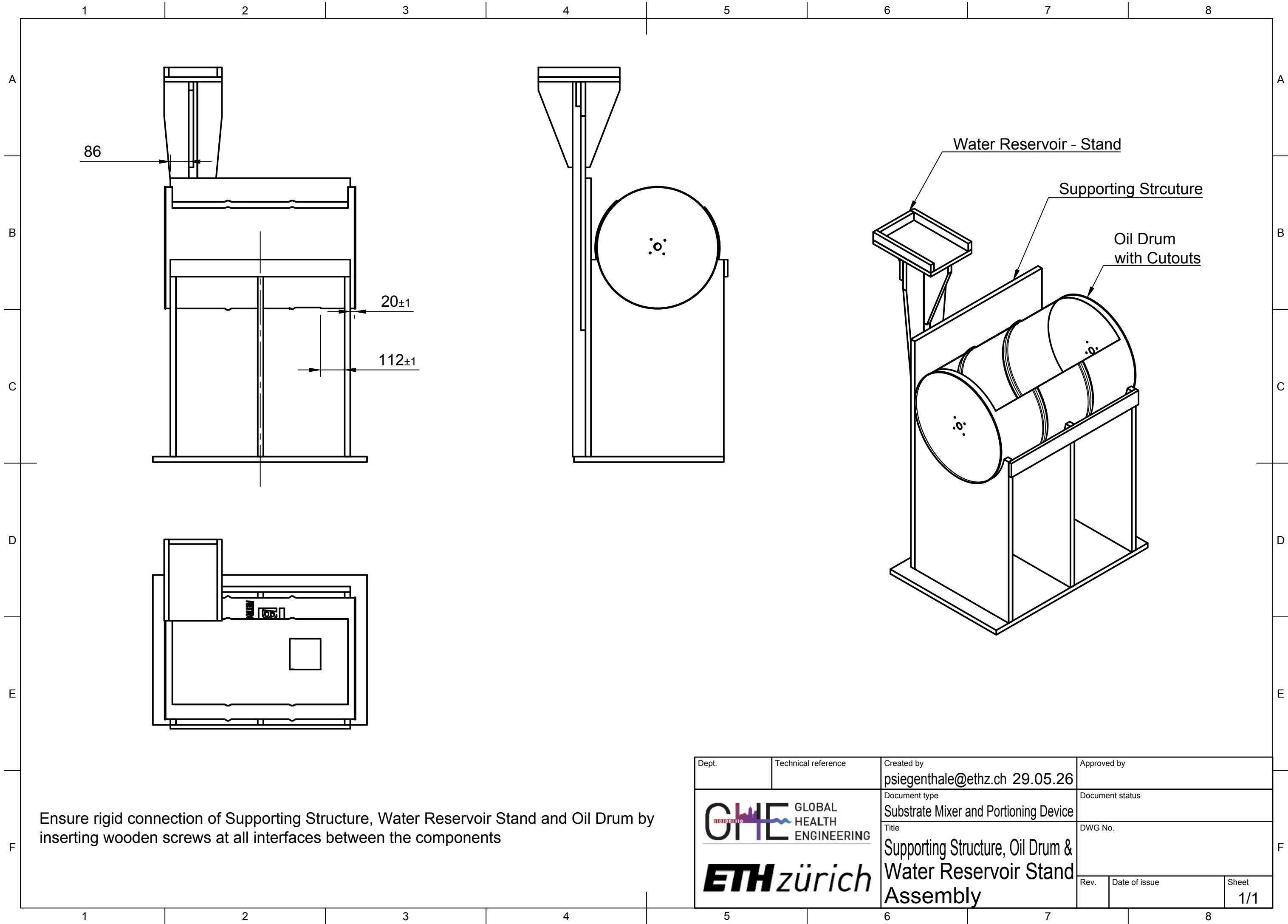


Use standardized 210 liter oil drum. ISO 15750

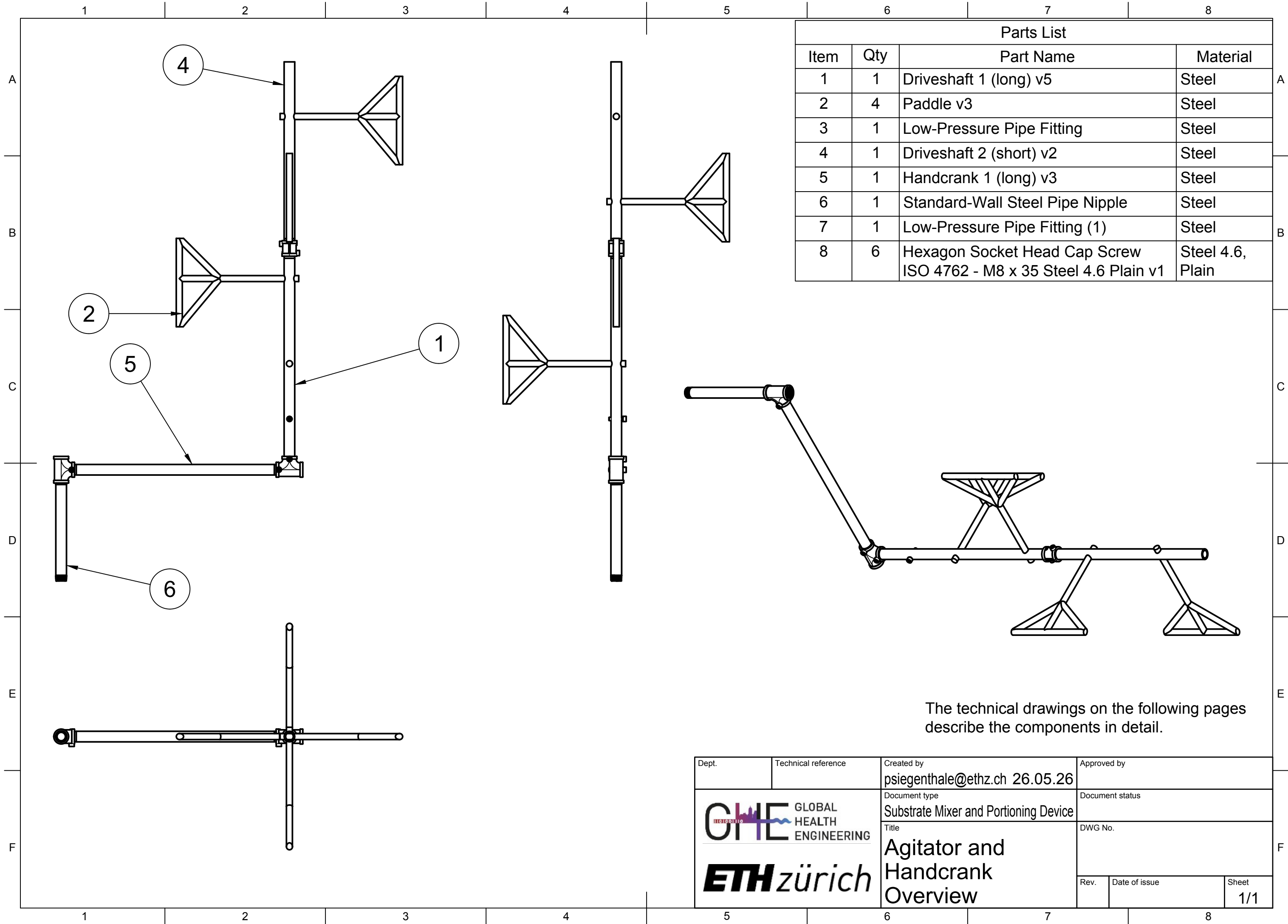
Cut out using angle grinder. Do not throw cutouts away. They are reused later as lid (upper, bigger cutout) and sealing plate (lower, smaller cutout)

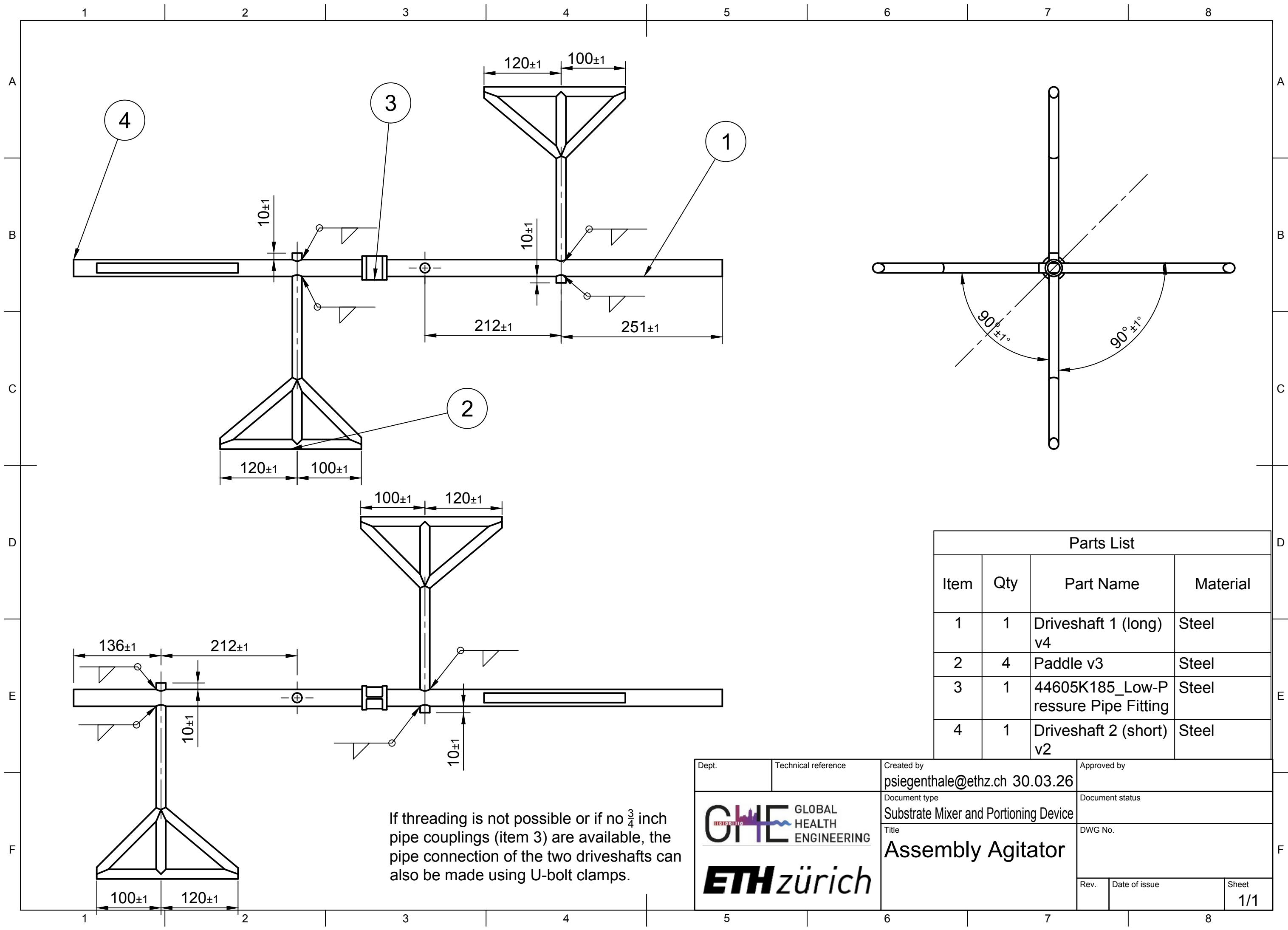
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	Title Mixing Vessel Oil Drum with Cutouts	DWG No.	
		Rev.	Date of issue
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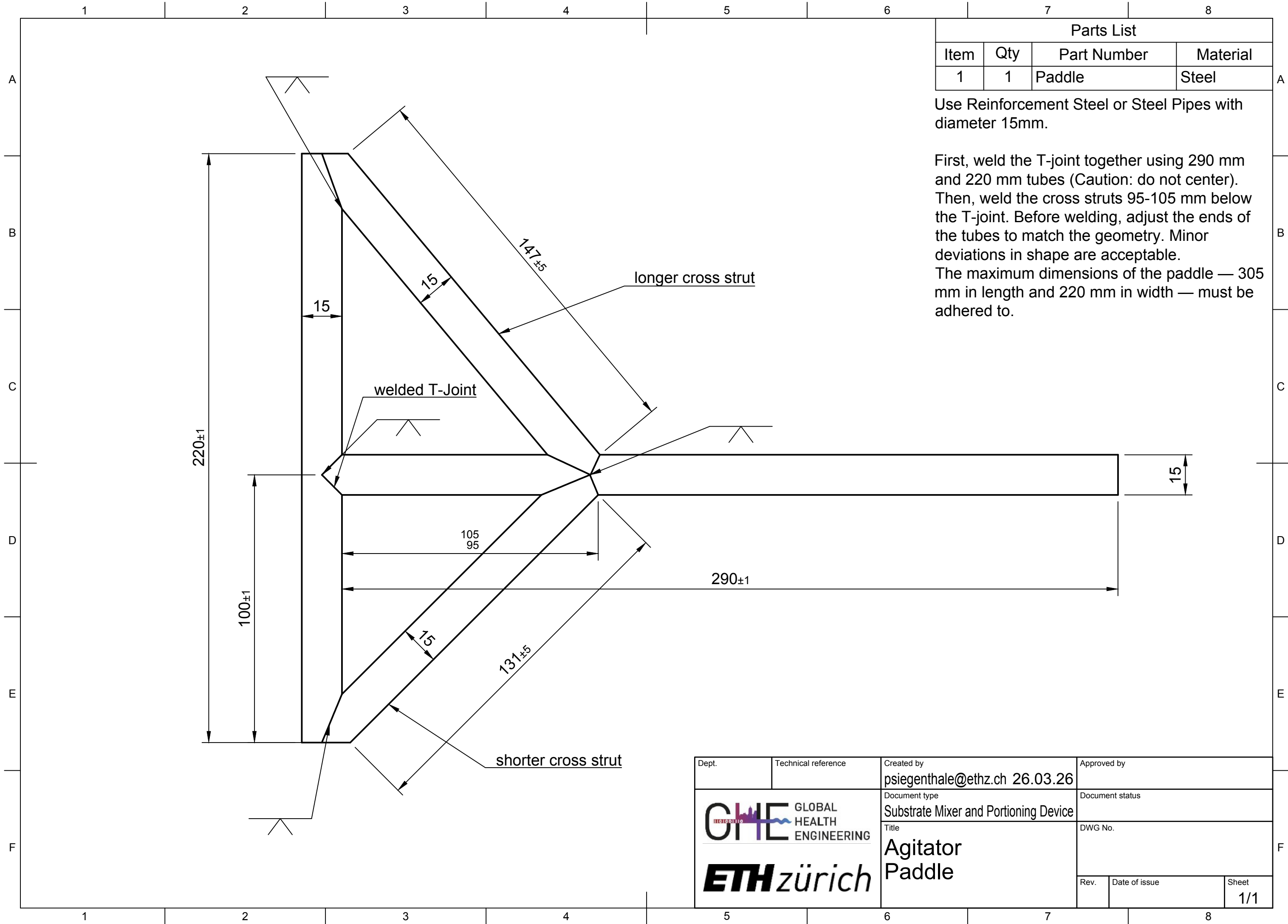


Agitator Components & Assembly

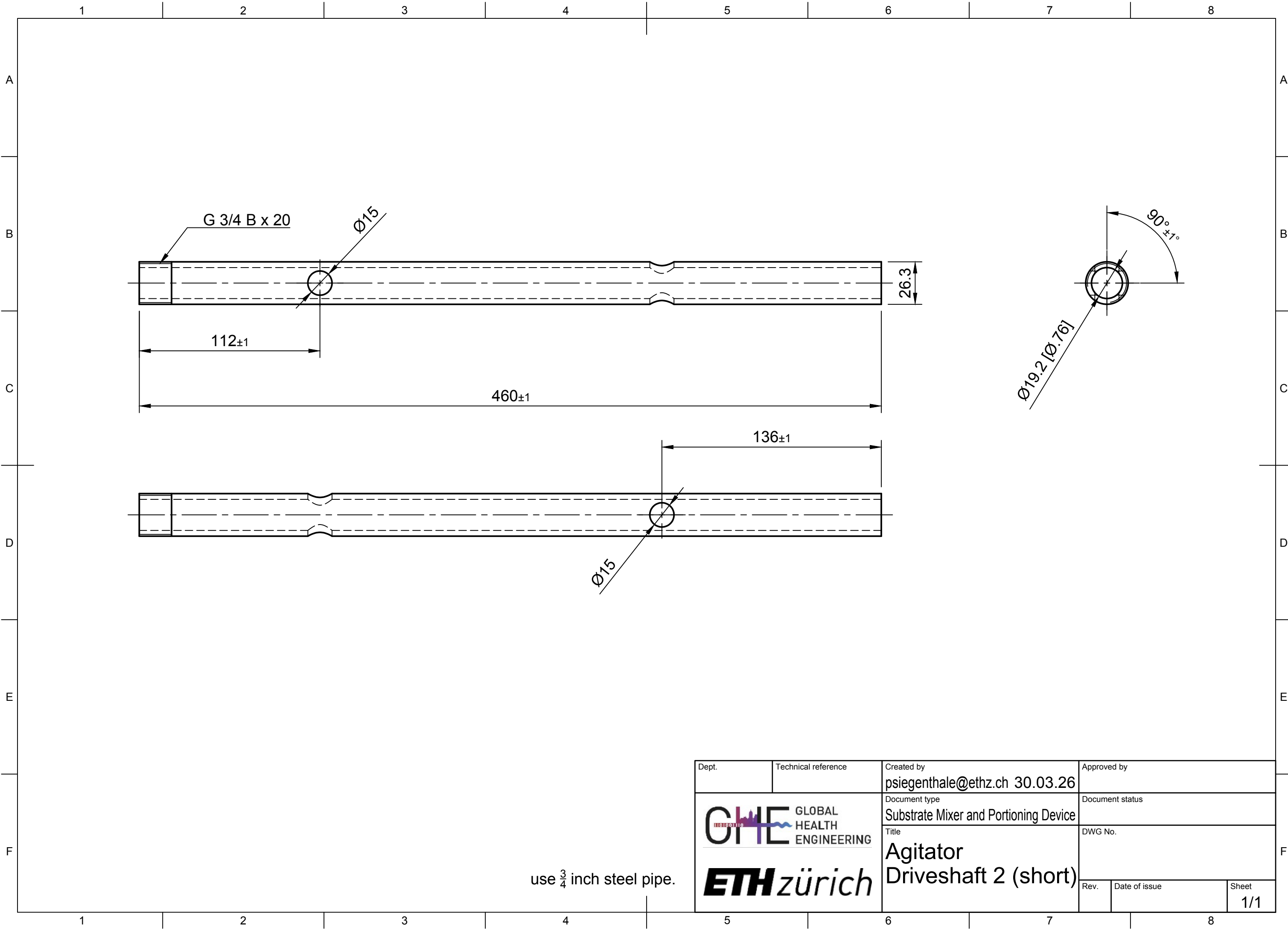




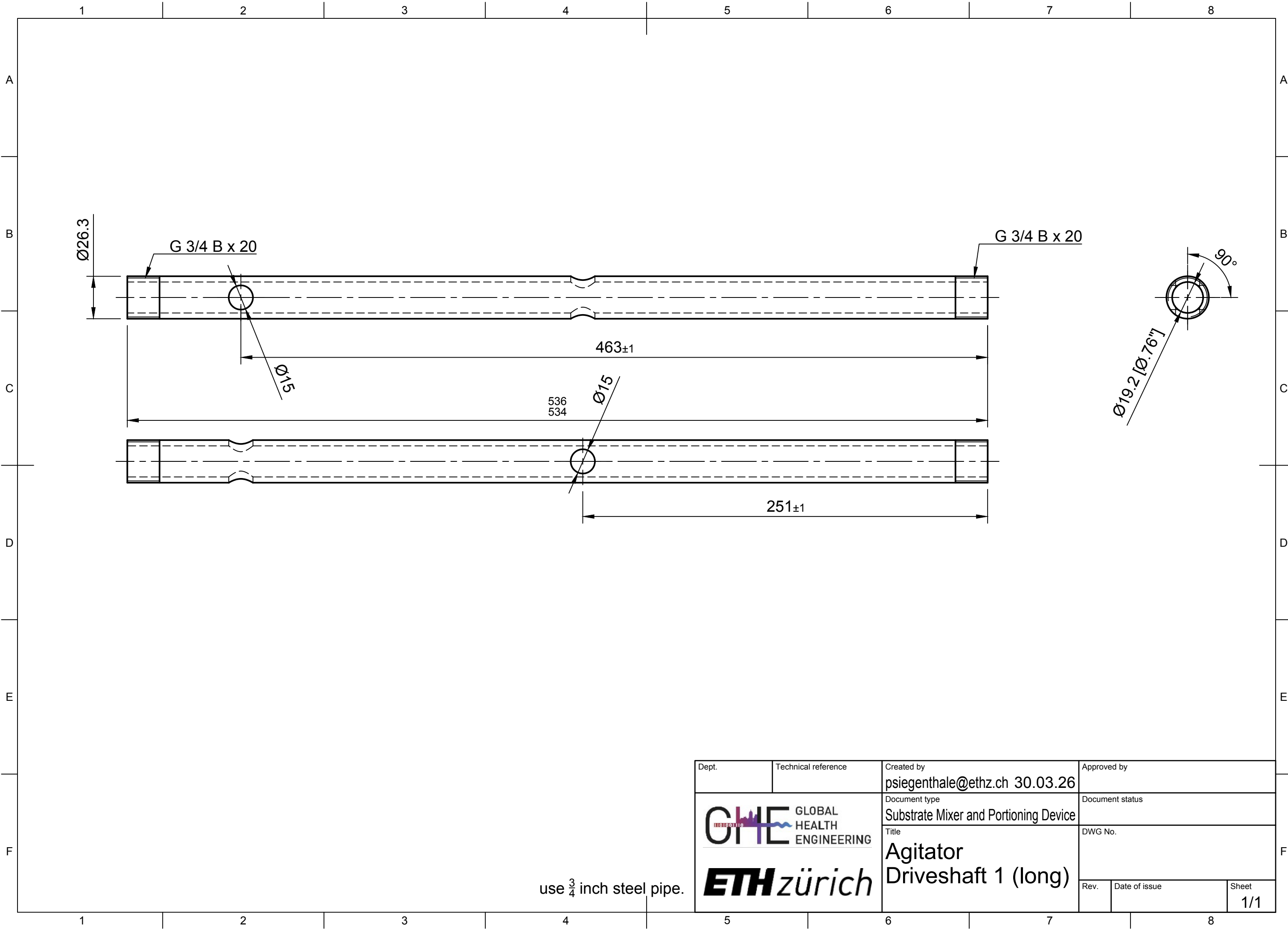
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 GLOBAL HEALTH ENGINEERING 		Document type Substrate Mixer and Portioning Device	Document status
		Title Assembly Agitator	DWG No.
		Rev.	Date of issue
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	Title Agitator Paddle		DWG No.
	Rev.	Date of issue	Sheet 1/1

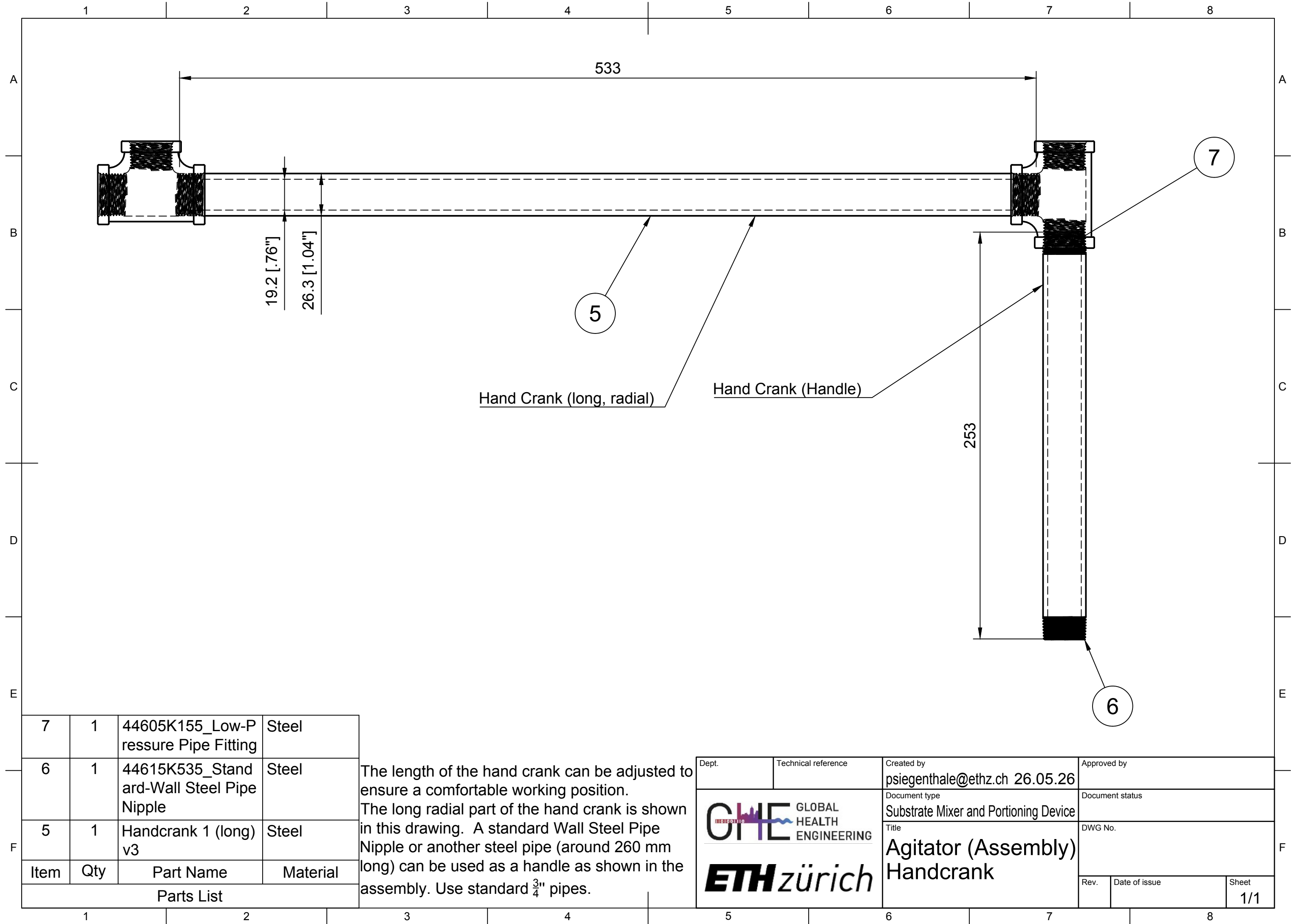


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		Title Agitator Driveshaft 2 (short)	DWG No.	
		Rev.	Date of issue	Sheet 1/1



use $\frac{3}{4}$ inch steel pipe.

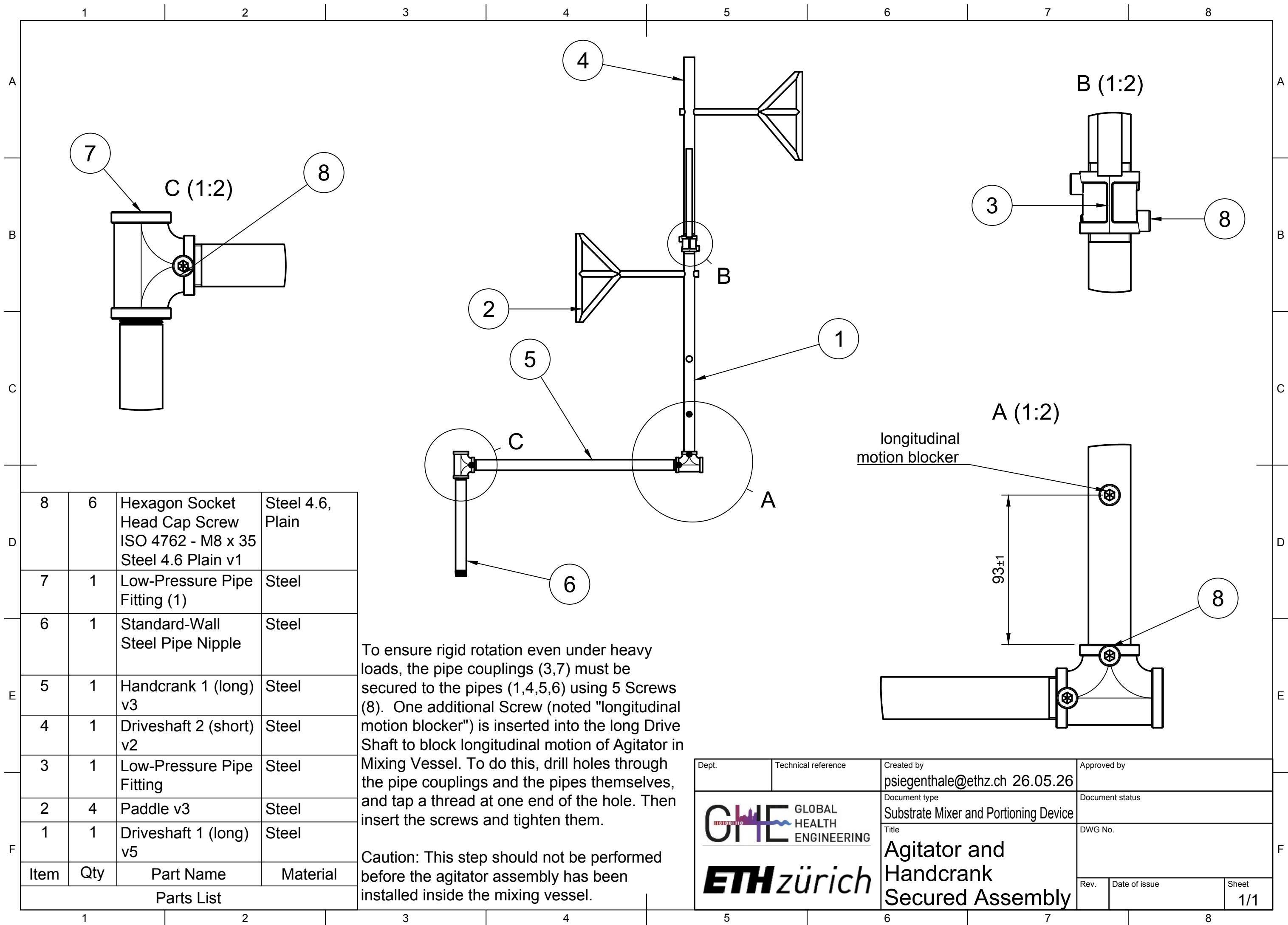
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		Title Agitator Driveshaft 1 (long)	DWG No.
		Rev.	Date of issue
		Sheet 1/1	

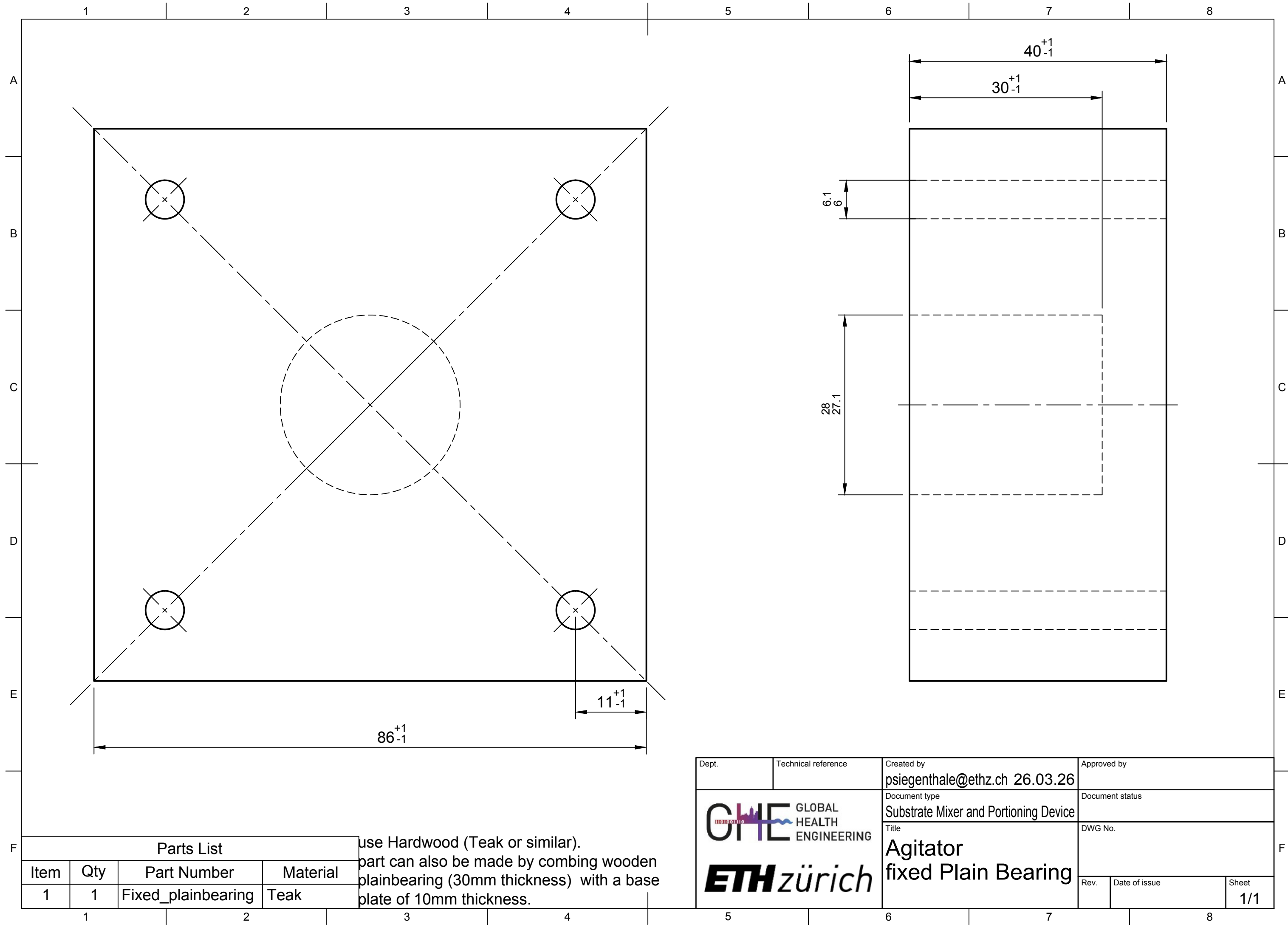


7	1	44605K155_Low-Pressure Pipe Fitting	Steel
6	1	44615K535_Standard-Wall Steel Pipe Nipple	Steel
5	1	Handcrank 1 (long) v3	Steel
Item	Qty	Part Name	Material
Parts List			

The length of the hand crank can be adjusted to ensure a comfortable working position. The long radial part of the hand crank is shown in this drawing. A standard Wall Steel Pipe Nipple or another steel pipe (around 260 mm long) can be used as a handle as shown in the assembly. Use standard $\frac{3}{4}$ " pipes.

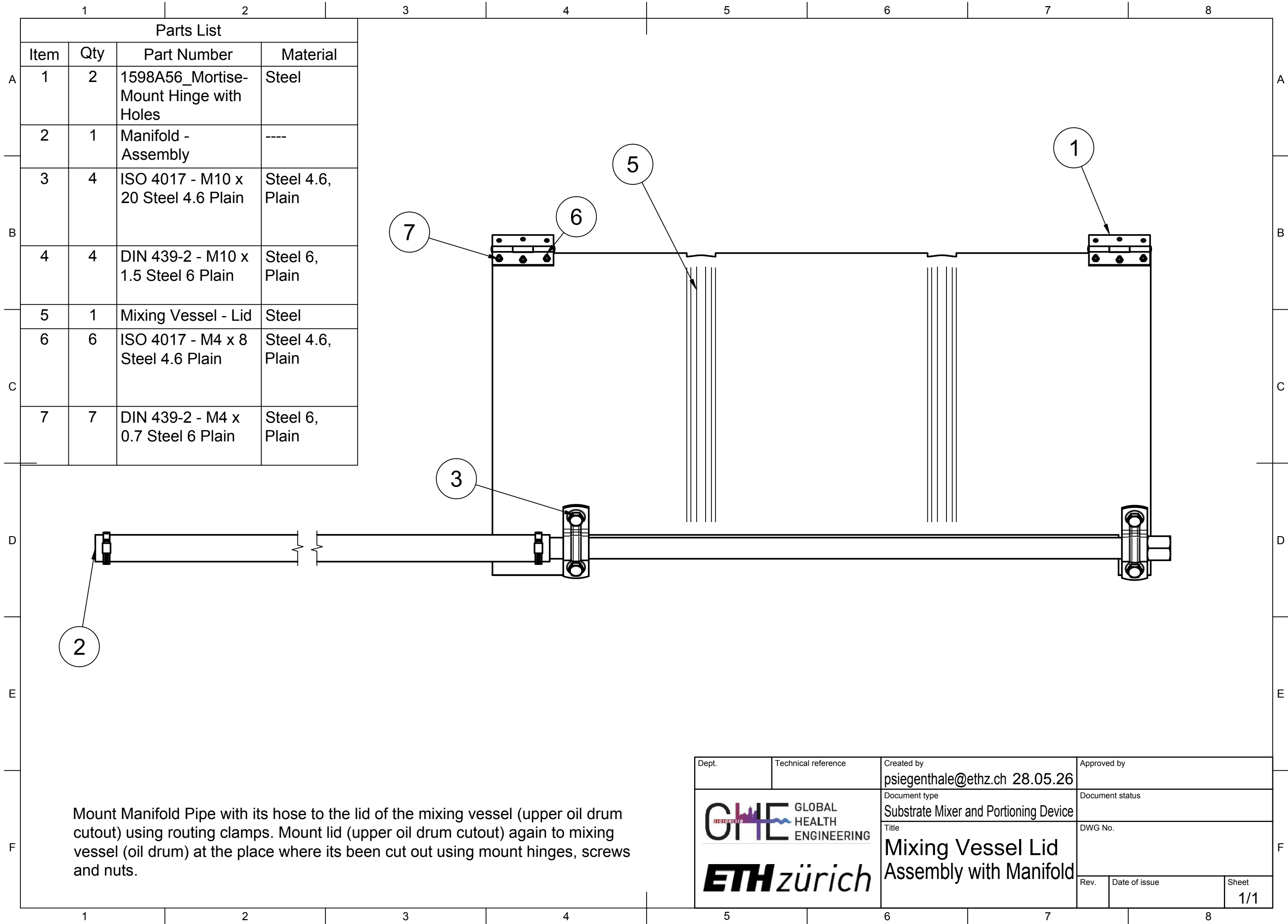
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		Title Agitator (Assembly) Handcrank	DWG No.
		Rev.	Date of issue
		Sheet	1/1

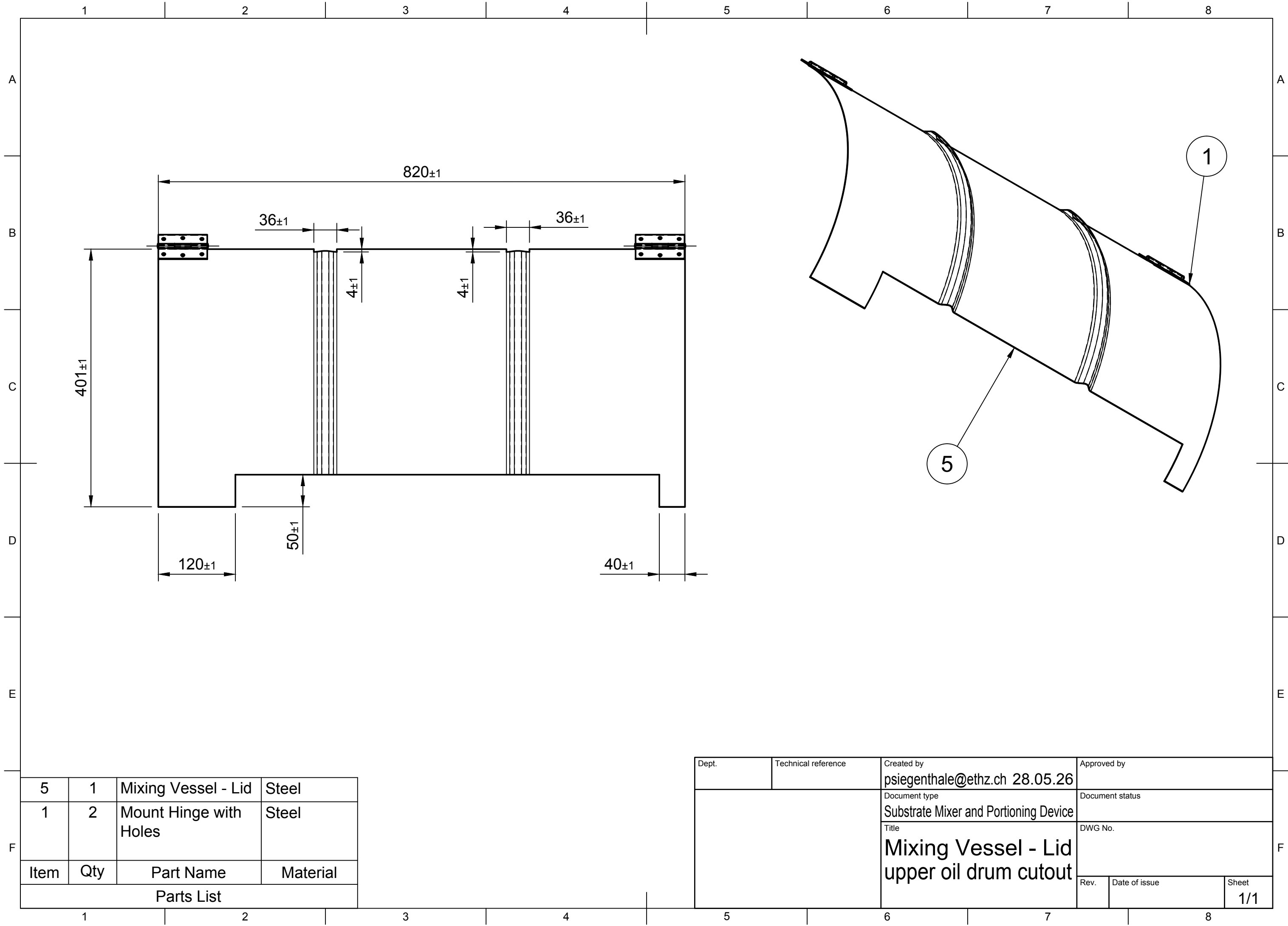




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		Document type Substrate Mixer and Portioning Device	Document status
		Title Agitator fixed Plain Bearing	DWG No.
		Rev.	Date of issue
		Sheet 1/1	

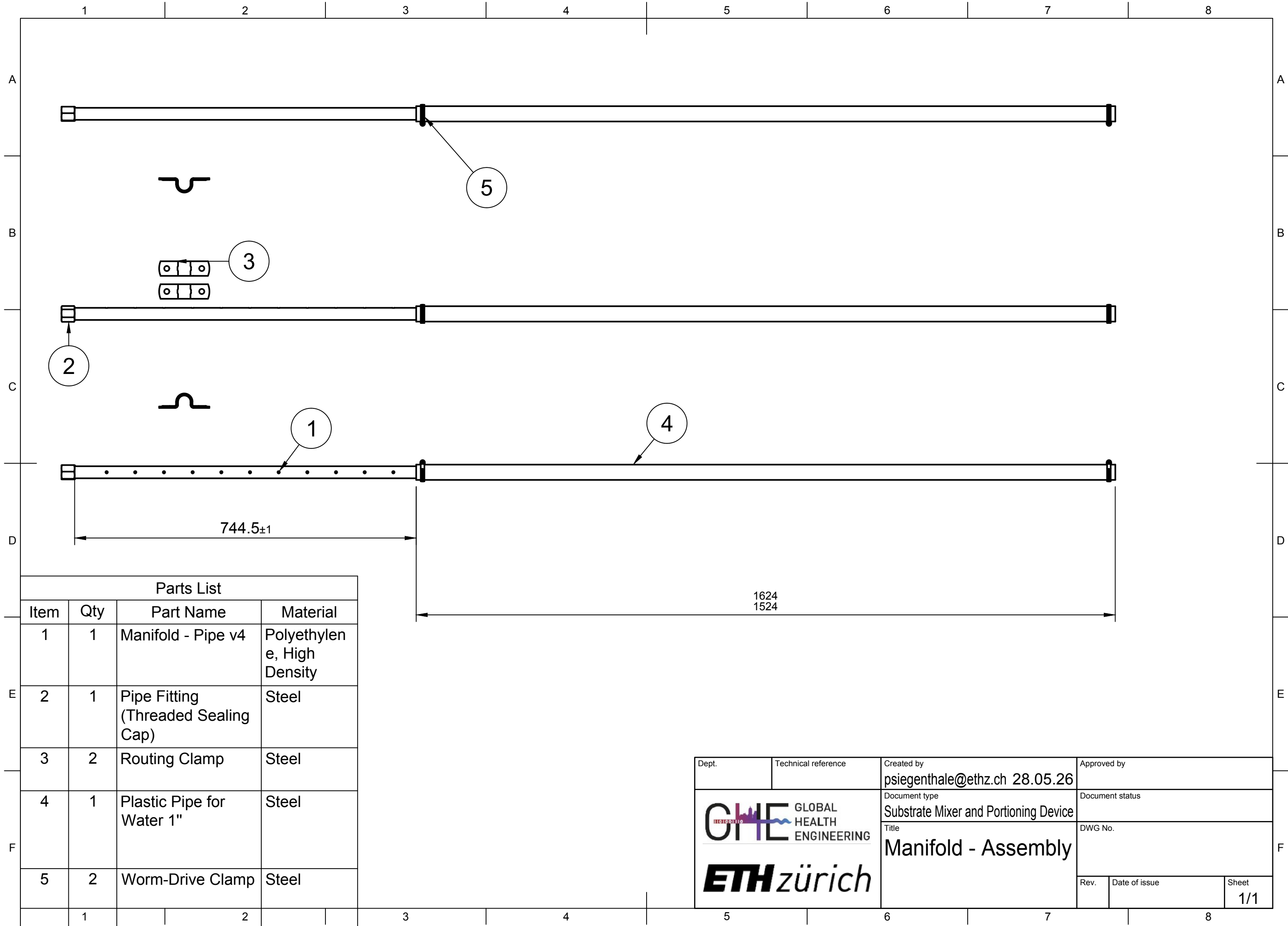
Water Manifold Components & Assembly



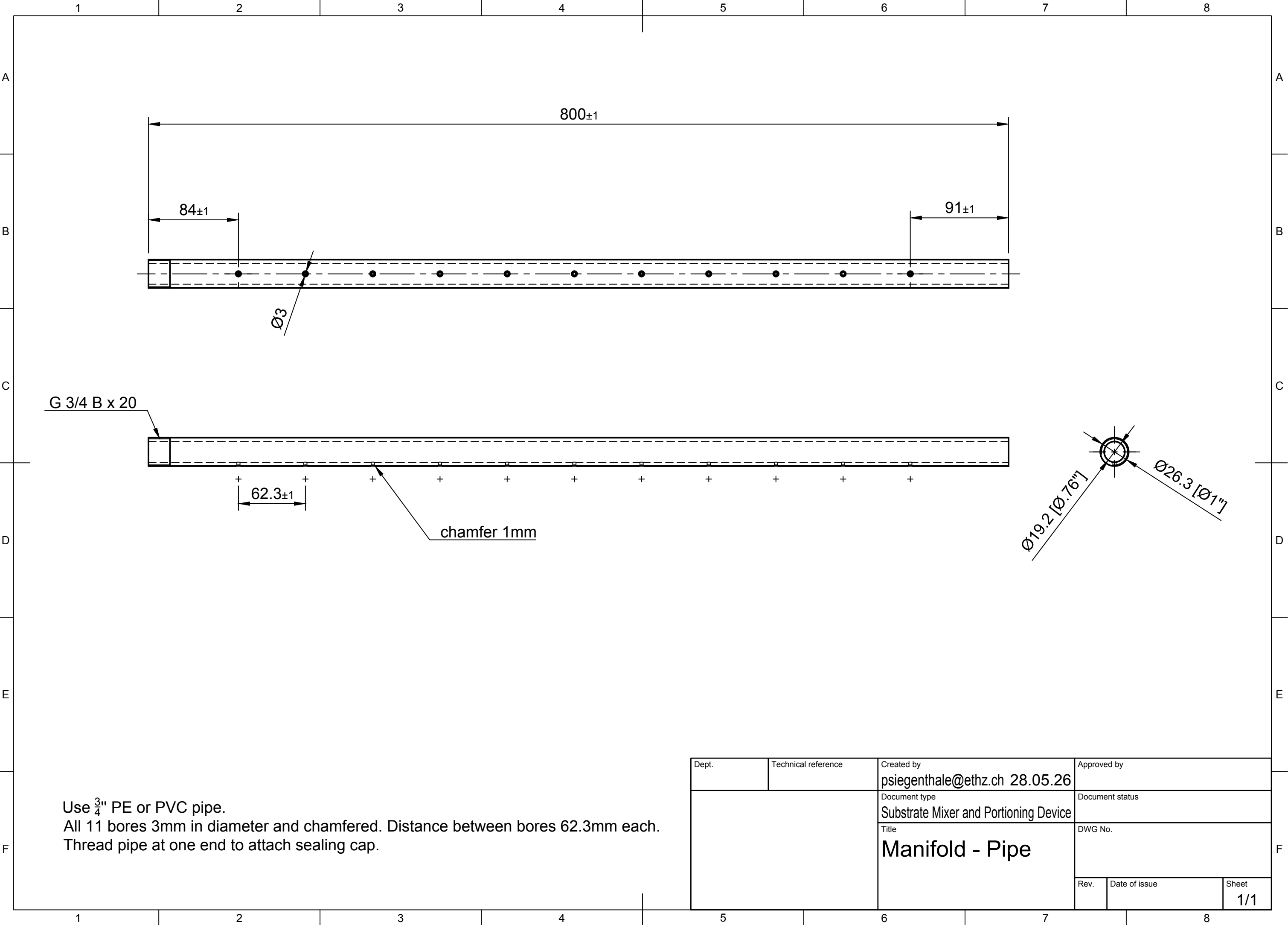


5	1	Mixing Vessel - Lid	Steel
1	2	Mount Hinge with Holes	Steel
Item	Qty	Part Name	Material
Parts List			

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		Document type Substrate Mixer and Portioning Device	Document status	
		Title Mixing Vessel - Lid upper oil drum cutout	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	

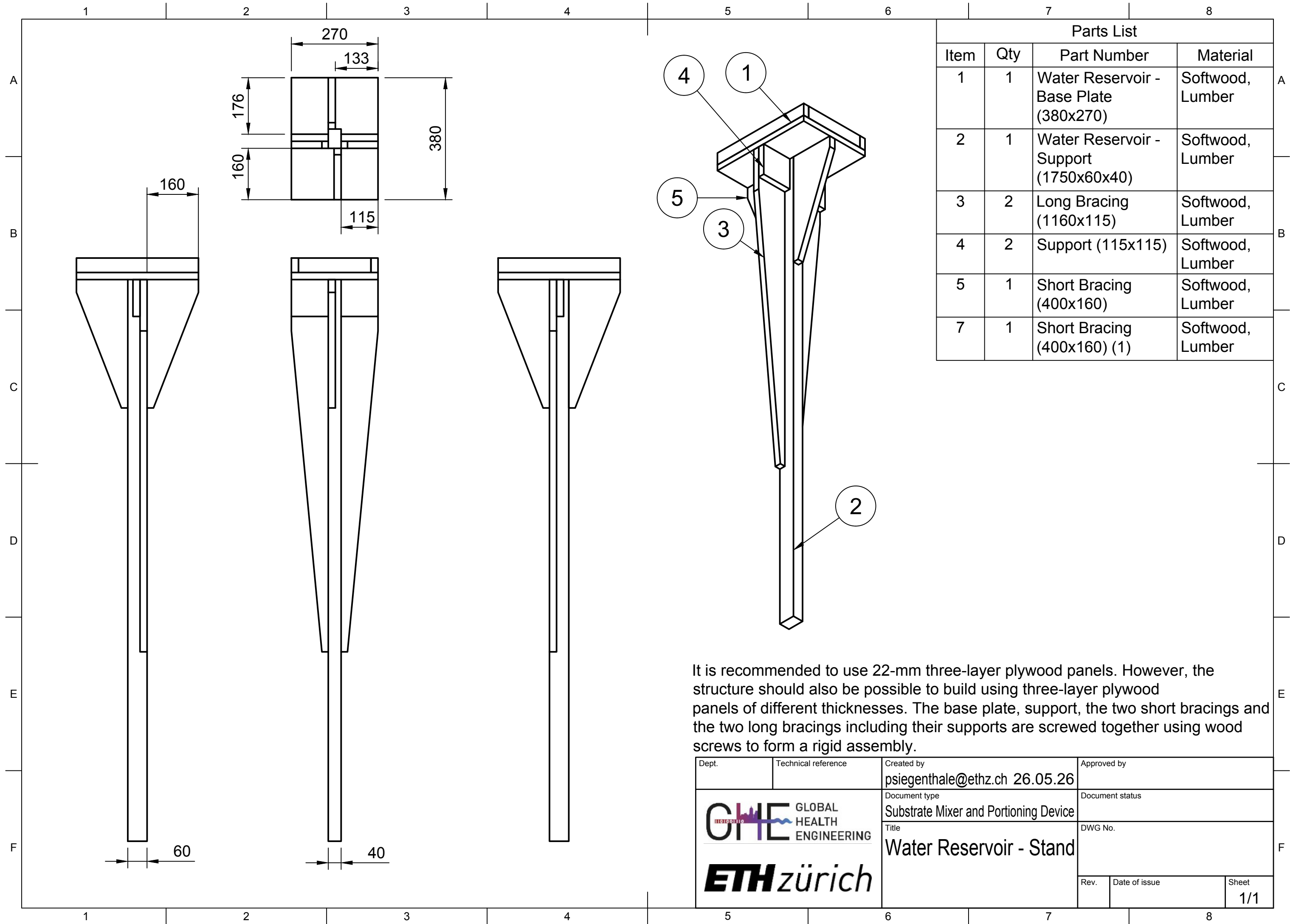


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		Title Manifold - Assembly	DWG No.
		Rev.	Date of issue
		Sheet 1/1	



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		Document type Substrate Mixer and Portioning Device	Document status	
		Title Manifold - Pipe	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	

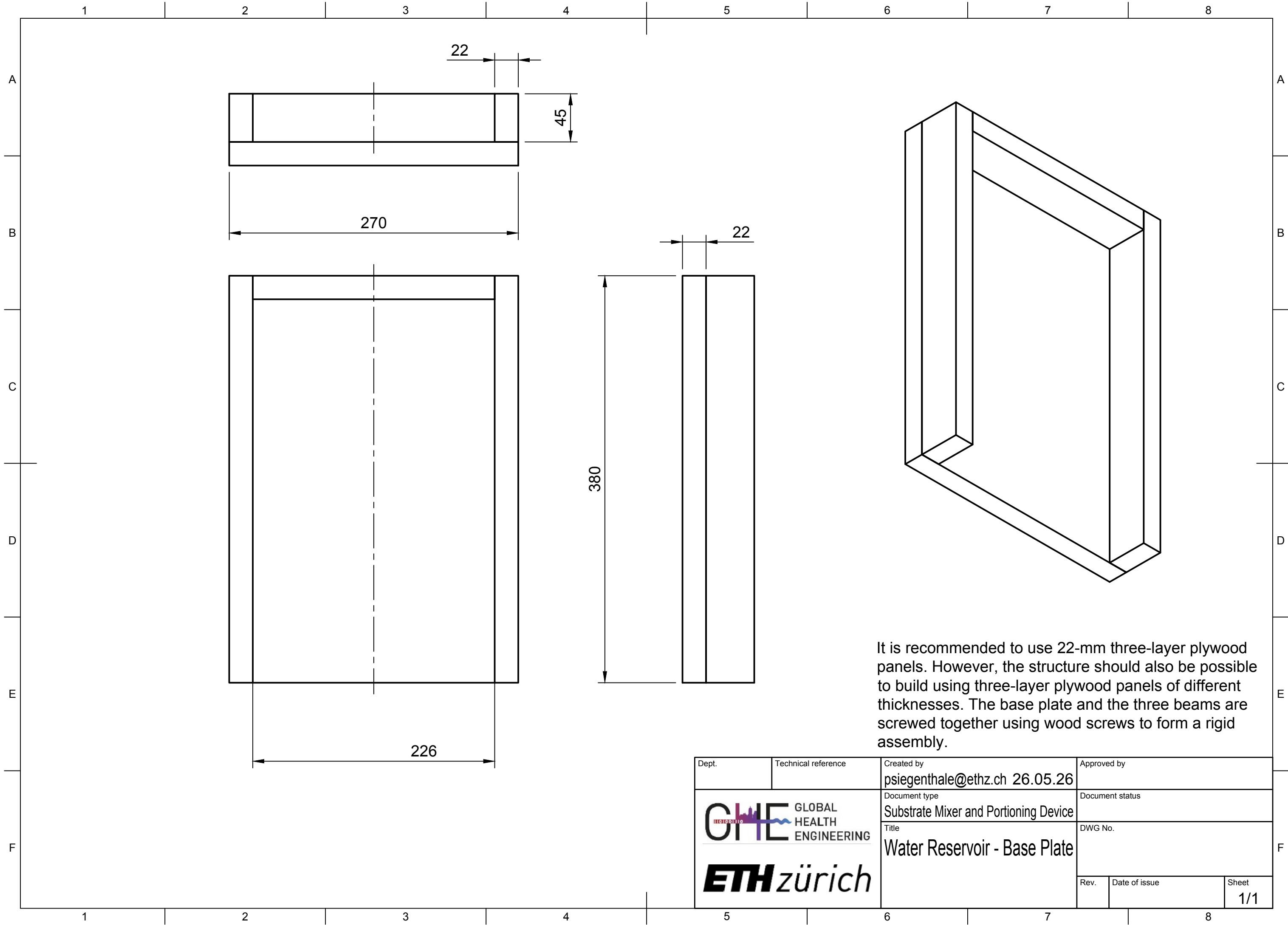
Water Reservoir Components & Assembly



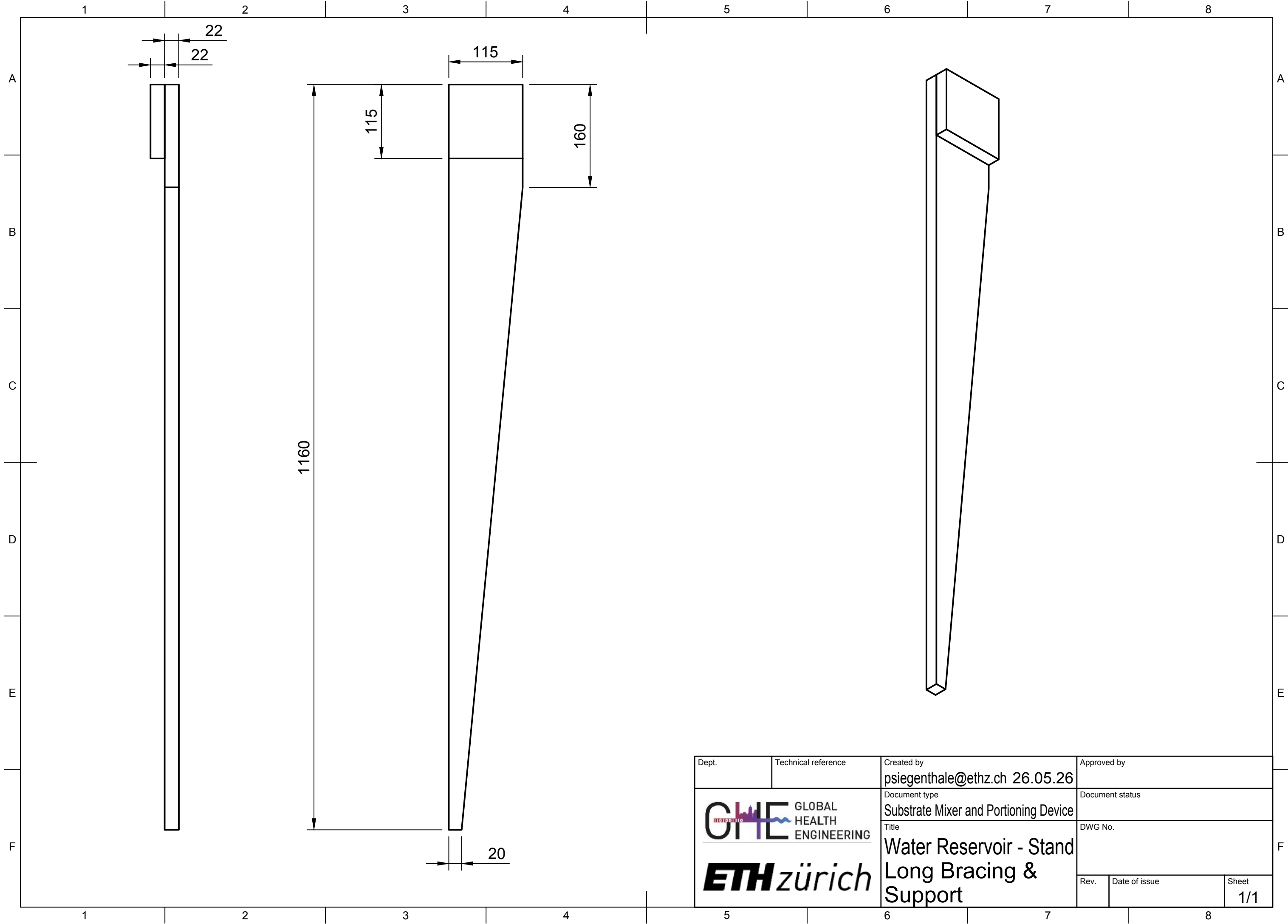
Parts List			
Item	Qty	Part Number	Material
1	1	Water Reservoir - Base Plate (380x270)	Softwood, Lumber
2	1	Water Reservoir - Support (1750x60x40)	Softwood, Lumber
3	2	Long Bracing (1160x115)	Softwood, Lumber
4	2	Support (115x115)	Softwood, Lumber
5	1	Short Bracing (400x160)	Softwood, Lumber
7	1	Short Bracing (400x160) (1)	Softwood, Lumber

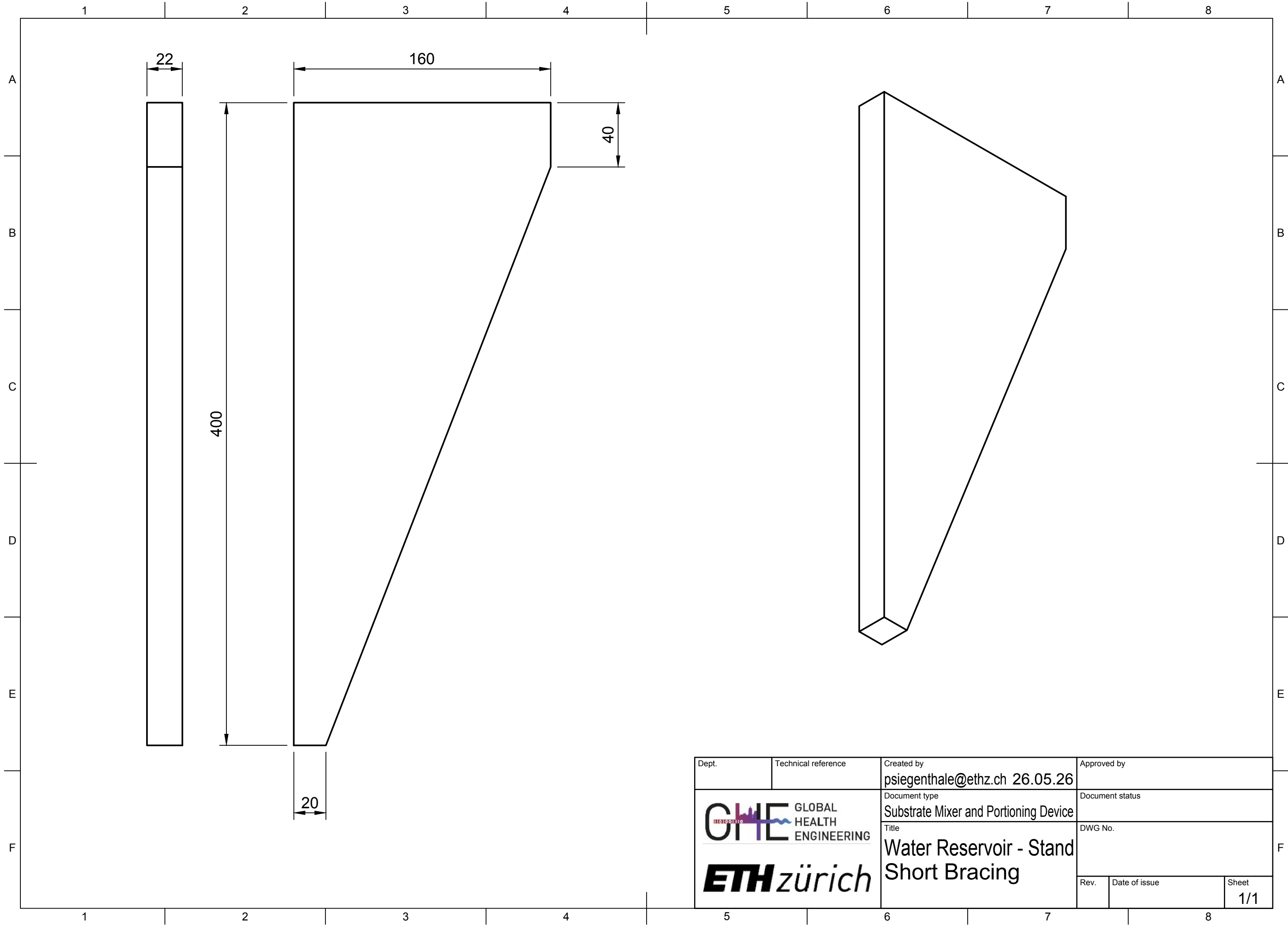
It is recommended to use 22-mm three-layer plywood panels. However, the structure should also be possible to build using three-layer plywood panels of different thicknesses. The base plate, support, the two short bracings and the two long bracings including their supports are screwed together using wood screws to form a rigid assembly.

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		Document type Substrate Mixer and Portioning Device	Document status
		Title Water Reservoir - Stand	DWG No.
		Rev.	Date of issue
		Sheet 1/1	

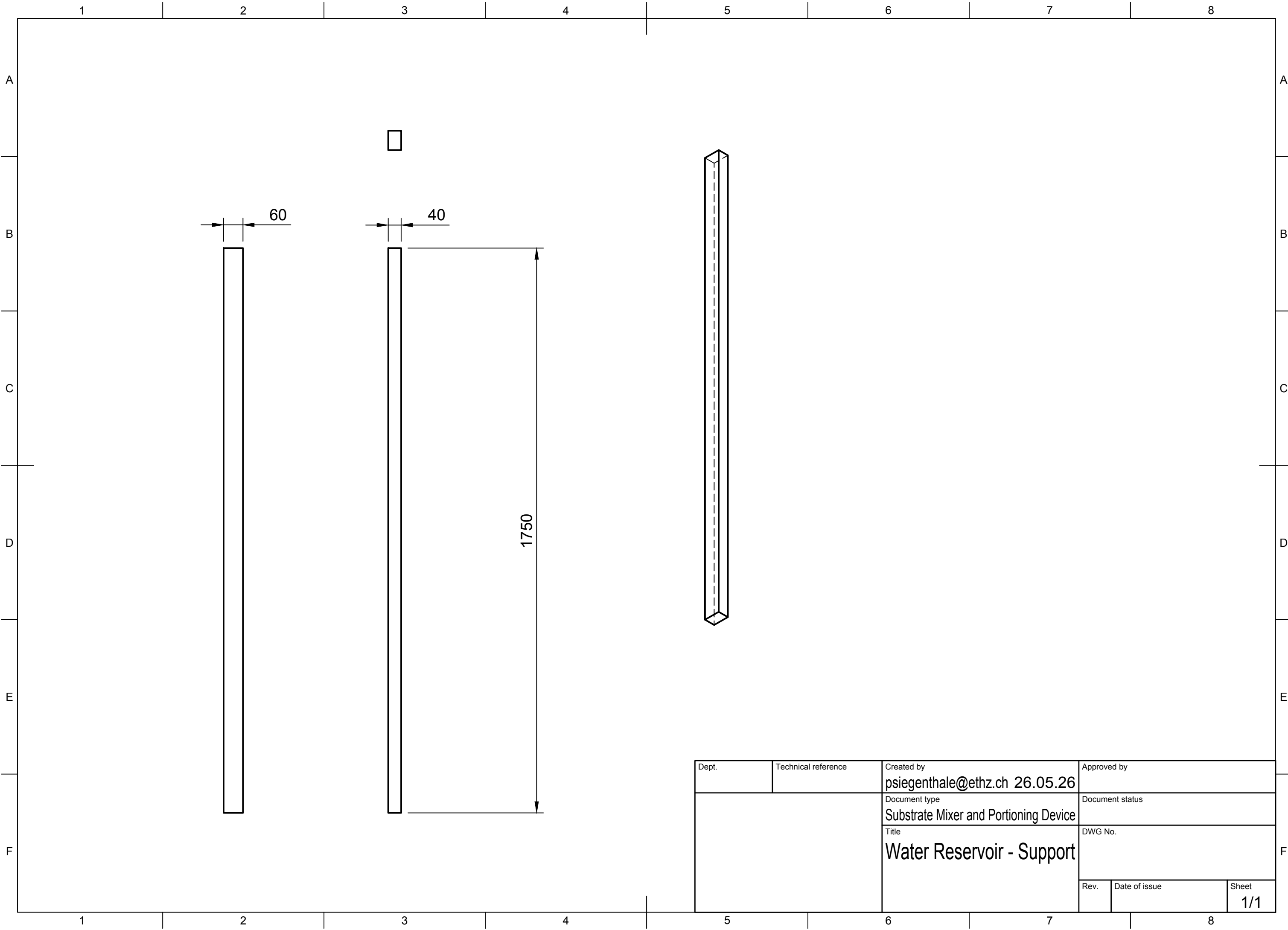


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	Title Water Reservoir - Base Plate	DWG No.	
	Rev.	Date of issue	Sheet 1/1

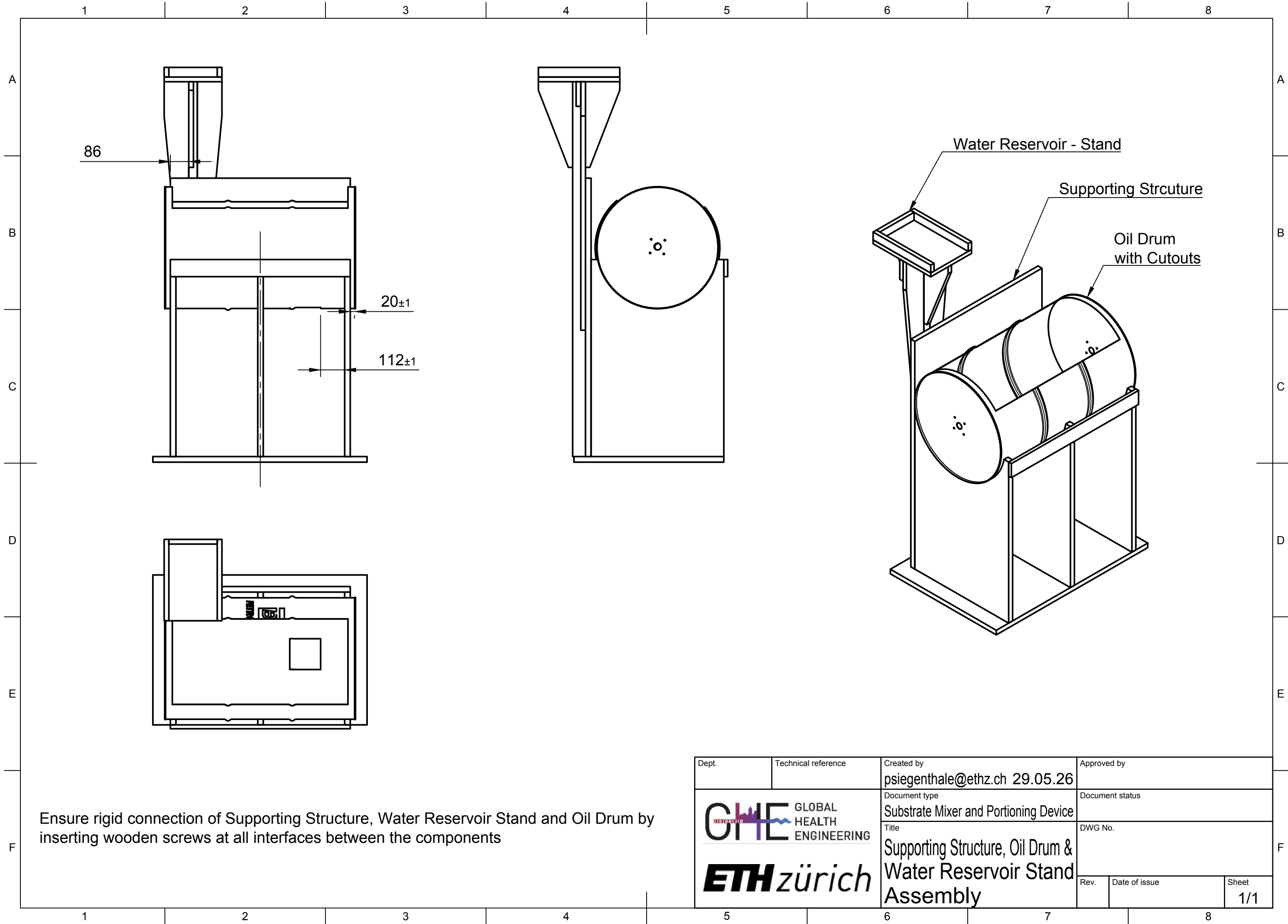




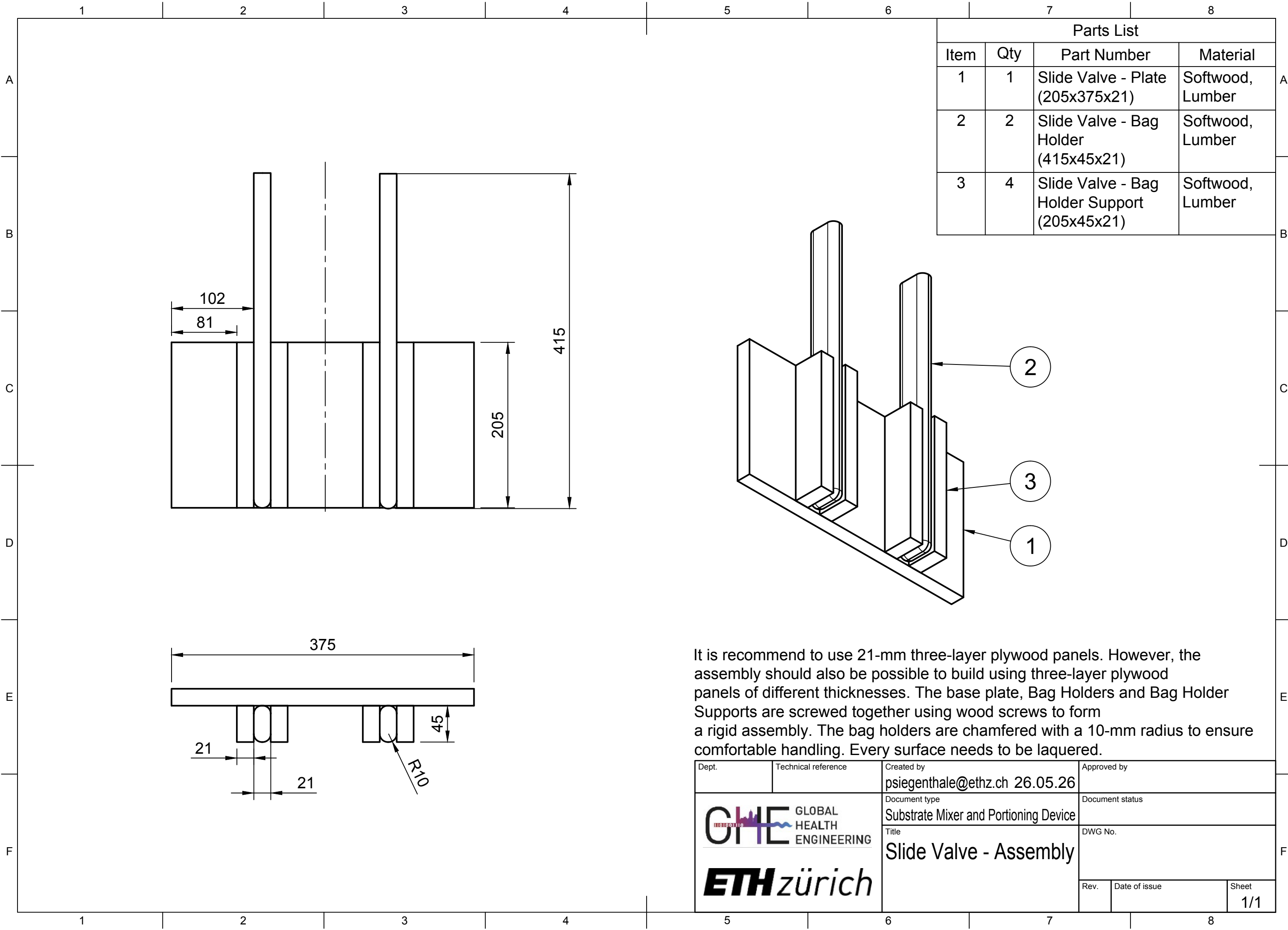
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	Title Water Reservoir - Stand Short Bracing	DWG No.	
	Rev.	Date of issue	Sheet 1/1



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		Document type Substrate Mixer and Portioning Device	Document status		
		Title Water Reservoir - Support	DWG No.		
			Rev.	Date of issue	Sheet 1/1

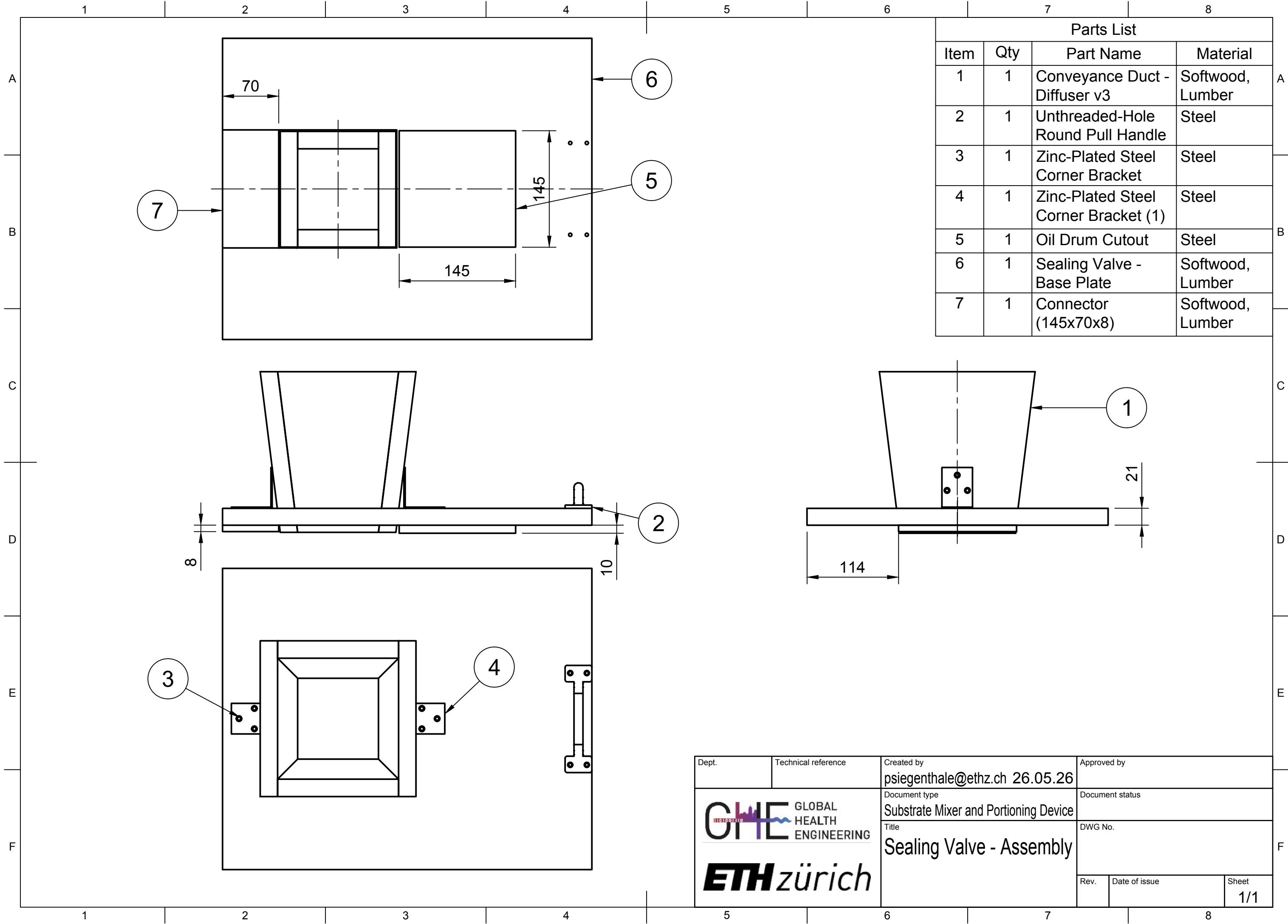


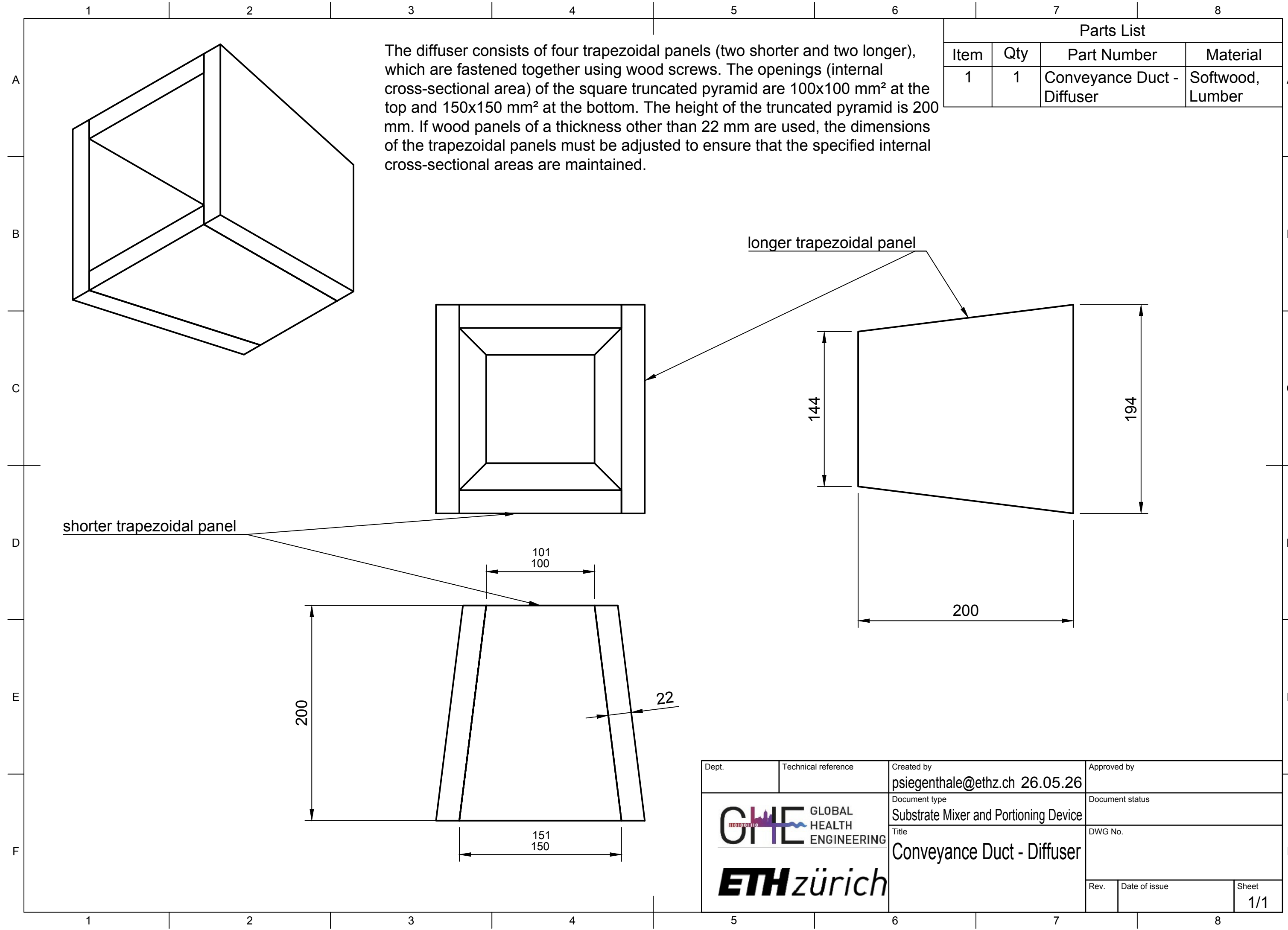
Filling Line Components & Assembly



It is recommend to use 21-mm three-layer plywood panels. However, the assembly should also be possible to build using three-layer plywood panels of different thicknesses. The base plate, Bag Holders and Bag Holder Supports are screwed together using wood screws to form a rigid assembly. The bag holders are chamfered with a 10-mm radius to ensure comfortable handling. Every surface needs to be laquered.

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	Document type Substrate Mixer and Portioning Device	Document status	
	Title Slide Valve - Assembly	DWG No.	
	Rev.	Date of issue	Sheet 1/1





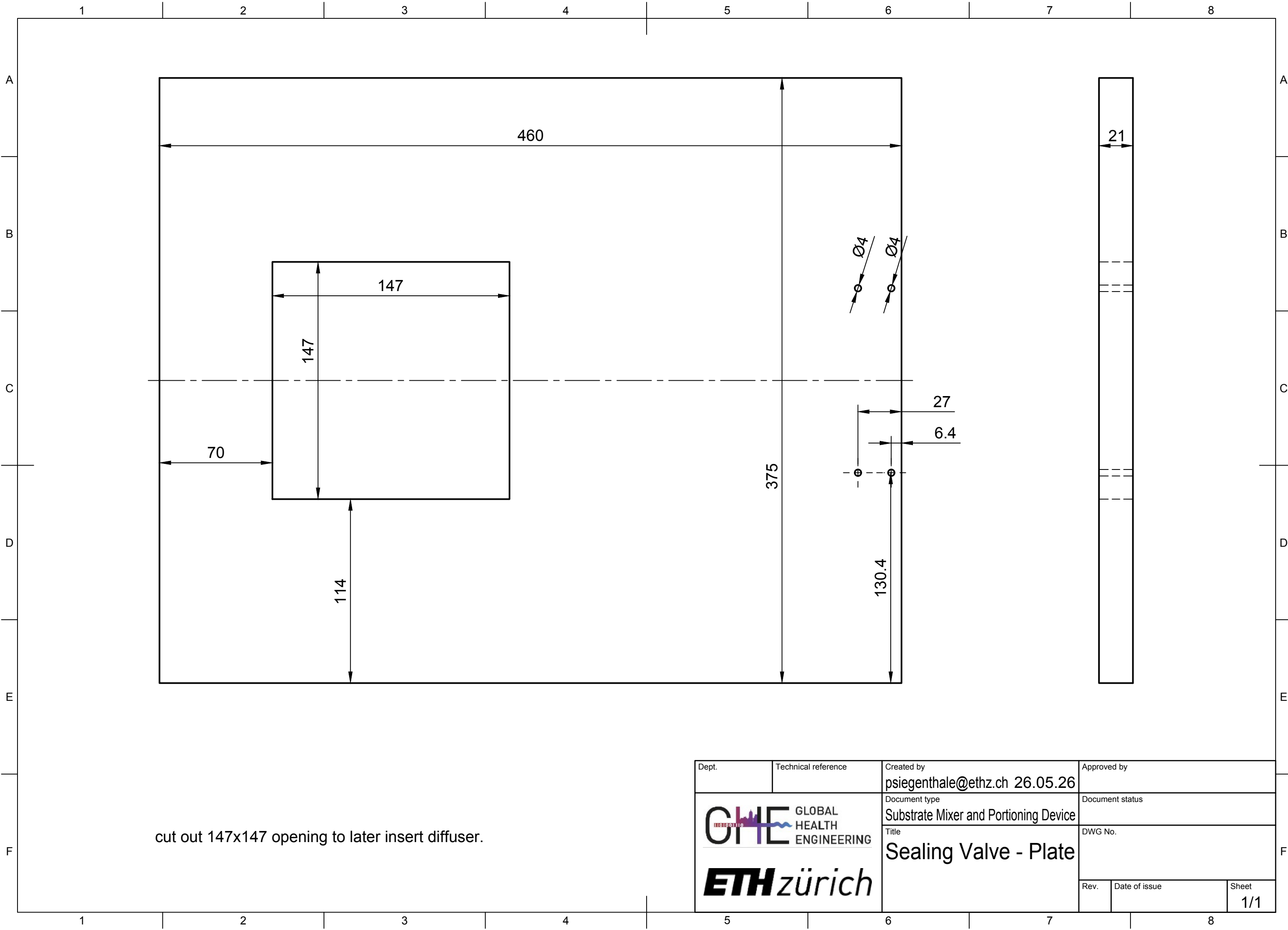
The diffuser consists of four trapezoidal panels (two shorter and two longer), which are fastened together using wood screws. The openings (internal cross-sectional area) of the square truncated pyramid are 100x100 mm² at the top and 150x150 mm² at the bottom. The height of the truncated pyramid is 200 mm. If wood panels of a thickness other than 22 mm are used, the dimensions of the trapezoidal panels must be adjusted to ensure that the specified internal cross-sectional areas are maintained.

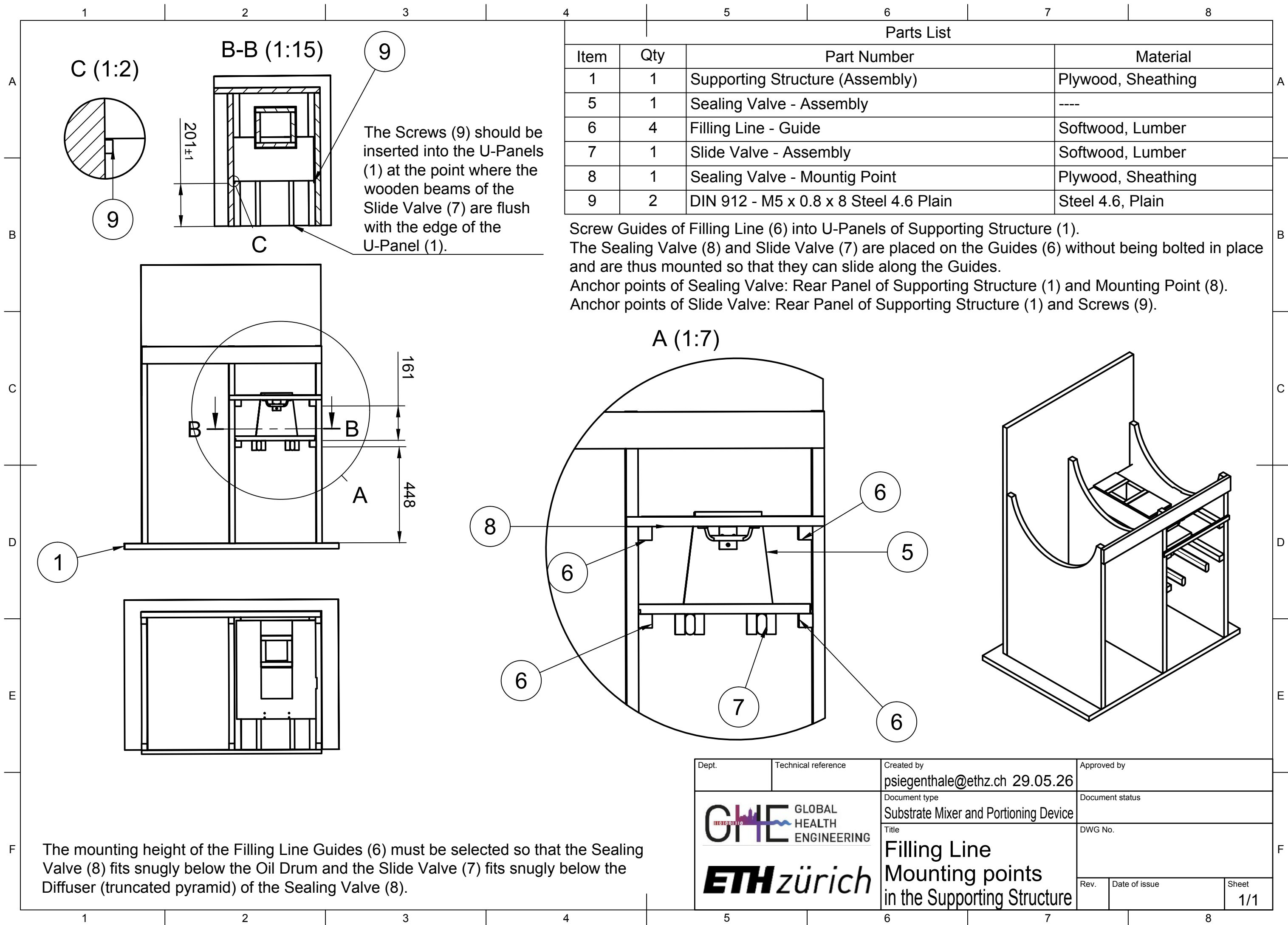
Parts List			
Item	Qty	Part Number	Material
1	1	Conveyance Duct - Diffuser	Softwood, Lumber

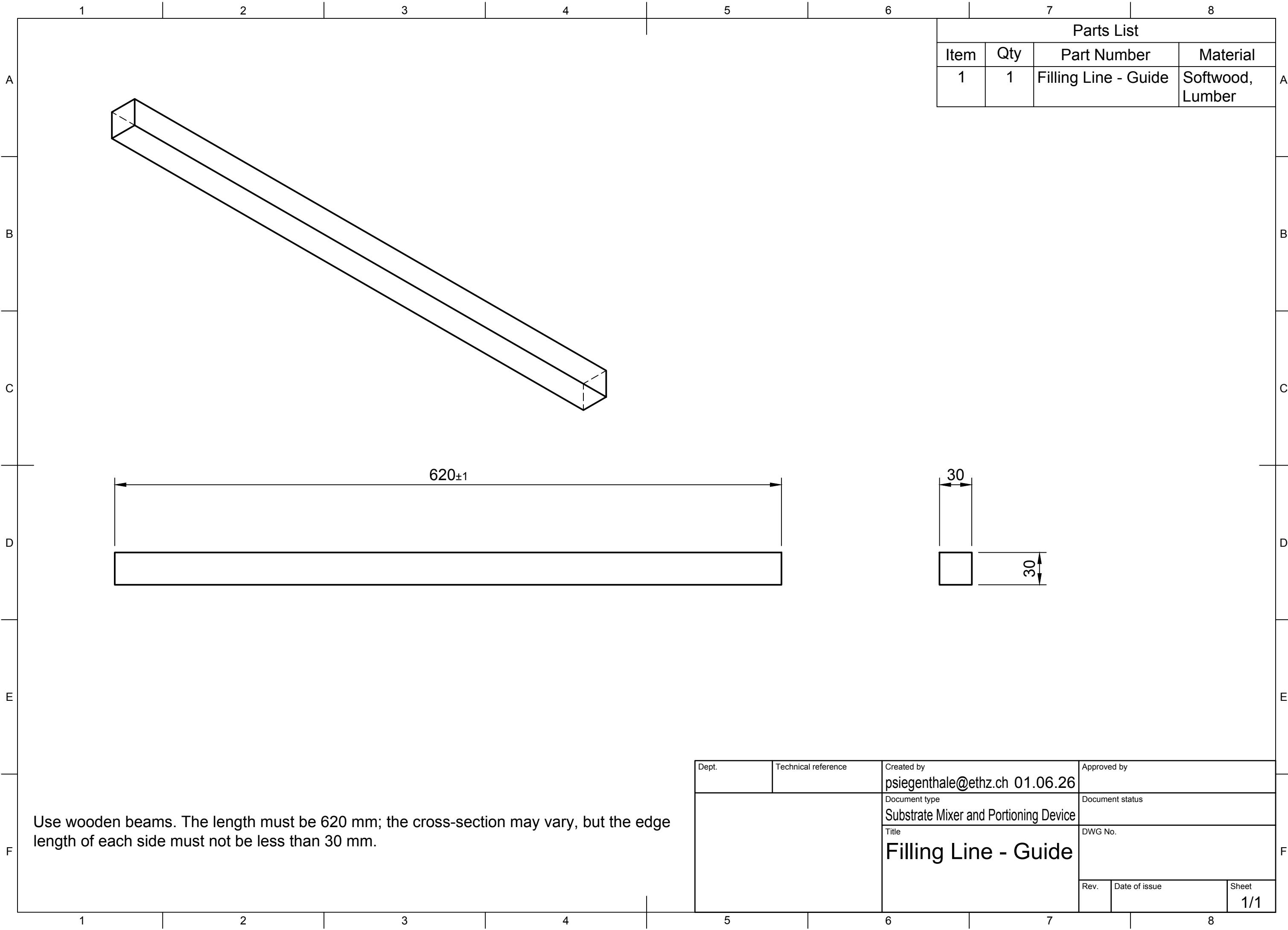
shorter trapezoidal panel

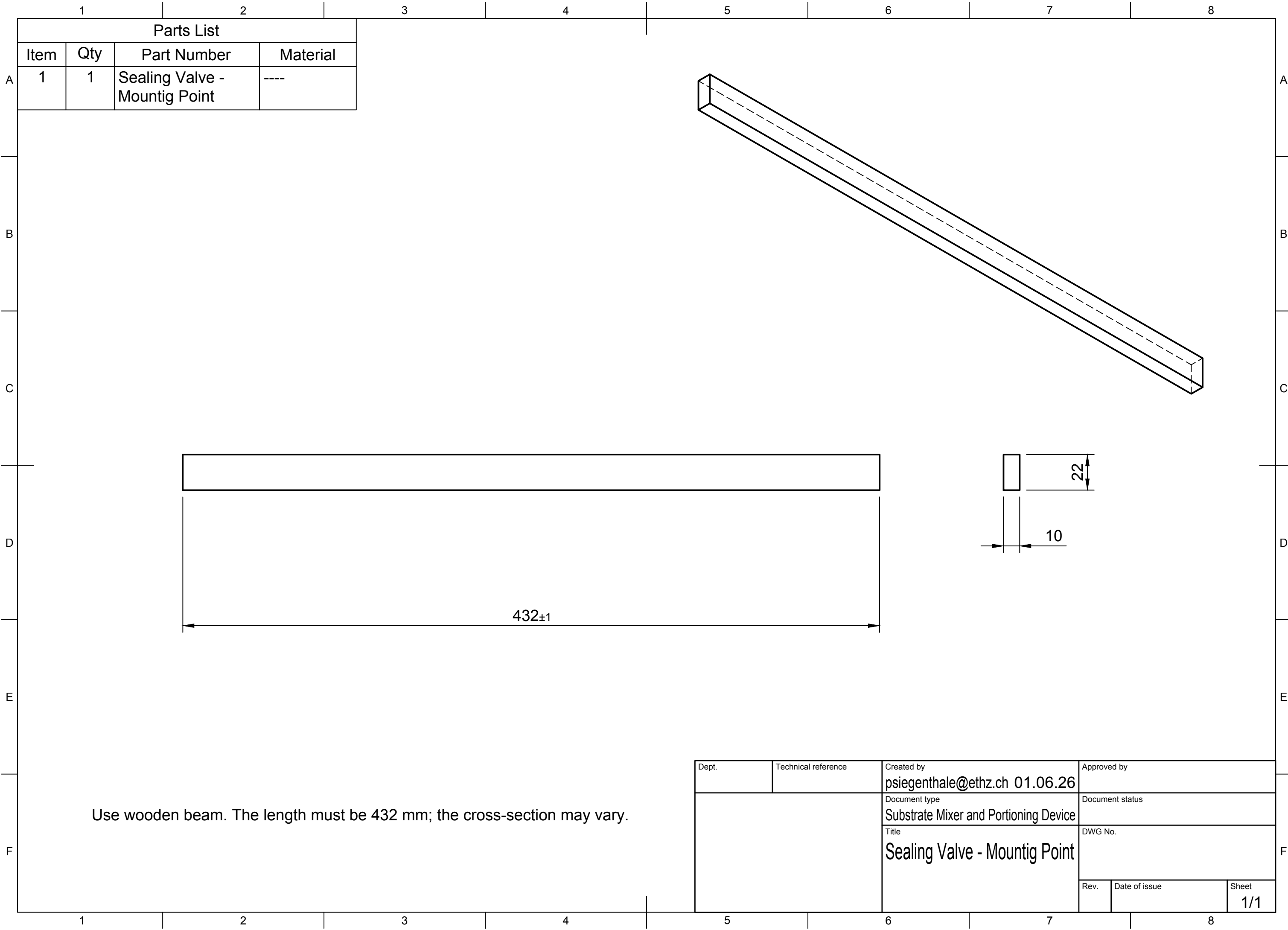
longer trapezoidal panel

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		Document type Substrate Mixer and Portioning Device	Document status
		Title Conveyance Duct - Diffuser	DWG No.
		Rev.	Date of issue
		Sheet 1/1	









Parts List			
Item	Qty	Part Number	Material
1	1	Sealing Valve - Mounting Point	----

Use wooden beam. The length must be 432 mm; the cross-section may vary.

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		Document type Substrate Mixer and Portioning Device	Document status	
		Title Sealing Valve - Mounting Point	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	

Assembly Instructions – Photo Documentation

Supporting Structure



Full view of the supporting structure with an unprocessed oil drum.



Lower Cutout. Oil drum laying on supporting structure for simpler cutting.

Oil Drum - Cutouts



Lower Cutout. Transition to the filling line. Without mounted sealing valve.



Upper, big Cutout. Oil drum laying on supporting structure. Cut-out piece is reused later as lid.

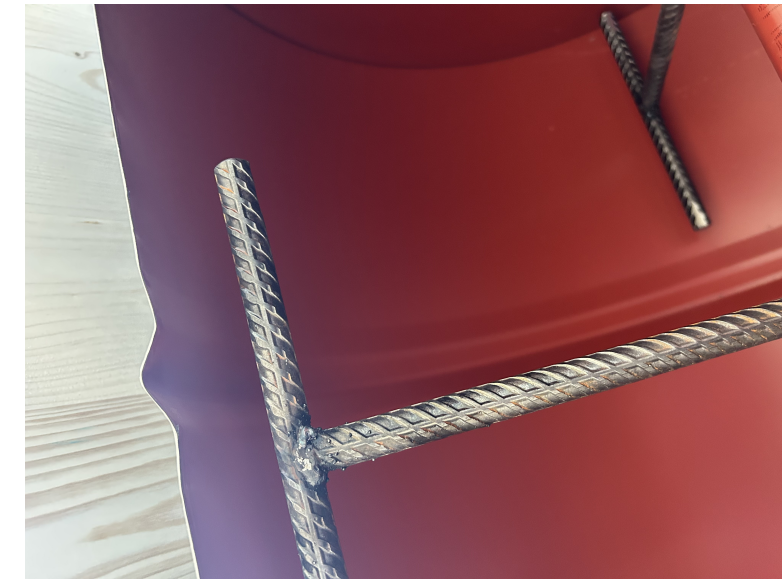
Agitator Components & Assembly



Full view of agitator assembly installed in oil drum.



The two welded components of the agitator before being installed and joined using pipe couplings and screws.



Welding process of paddle: T-piece is welded first. Bars to create a V are added later.



Welding process: Prefabricated, final V-paddles made of welded-together reinforcing bars (rebar).



Welding process: Weld rebar V-Paddle and iron pipe together.



Pipe couplings: To ensure rigid rotation even under heavy loads, the pipe couplings must be secured to the pipes using screws in threaded holes.



Welding process: Connecting the paddle to the driveshaft



Pipe coupling of hand crank and wooden bearing used to support the drive shaft.



The wooden slide bearing is bolted to the oil drum.



Water manifold (sprinkler): Water flows out of drilled and chamfered 3mm-holes. Work carefully to avoid creating ridges in the drill hole. Picture shows a bad example.

Water Manifold Components & Assembly



Full view of the water manifold with its lid (big oil drum cutout).



The water manifold is used as a stop for the lid on the oil drum. One end of the manifold is sealed with a pipe cap.



At the other end of the manifold, the water hose is secured with a hose clamp.



The lid (oil drum cutout) is bolted to the oil drum using hinges.



The manifold is bolted to the lid using routing clamps.



The lid (oil drum cutout) is bolted to the oil drum using hinges.



The manifold is bolted to the lid using routing clamps.



A suitable drain valve is the connection between the jerrycan and the water hose.



A suitable drain valve is the connection between the jerrycan and the water hose.



The bracket is mounted flush with the supporting structure.

Water Reservoir Components & Assembly



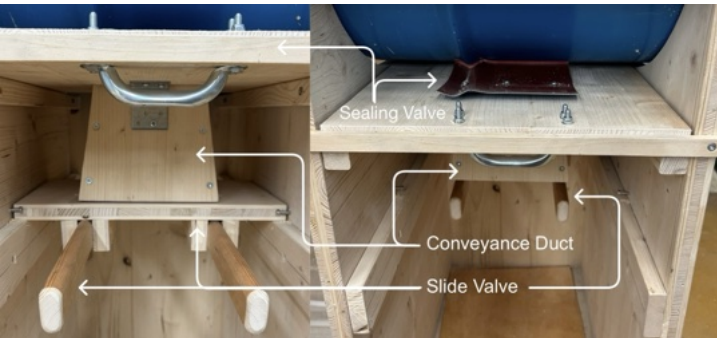
The stand and supporting structure are screwed together



The jerrycan is sitting on the plate, where braces support the weight. The square wooden boards make it easier to screw them in place.

Filling Line Components & Assembly

Filling Line - Assemblies and Guides



Full view of the filling line.



Full view of the Filling Line.



Filling Line. Valve guide rails.



Assembly of Filling Line. Sealing Valve with red oil drum cutout in open position.

Filling Line - Sealing Valve



Assembly of Filling Line. Slide valve below diffuser (truncated pyramid) in closed position.



Sealing valve top view.



Sealing valve side view.



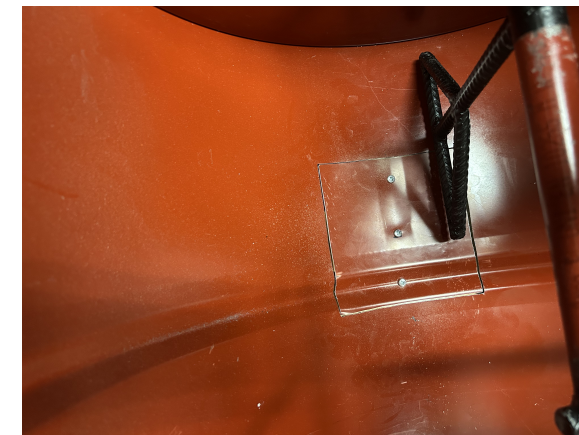
Sealing valve bottom view.



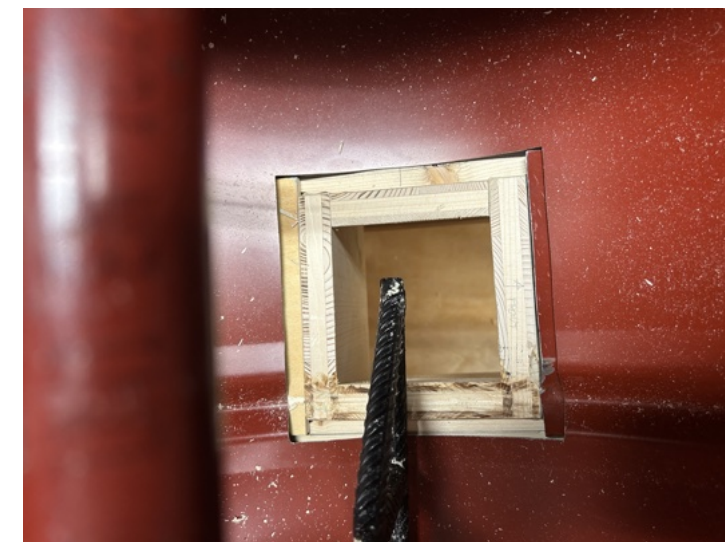
The red oil drum cutout in front of the diffuser (truncated pyramid).



The sealing valve plate must be trimmed (using an angle grinder or similar tool) so that the oil drum cutout can be installed flush.



Sealing valve in closed position. The cutout (mounted on the sealing valve plate) is flush with the oil drum, filling out the oil drum's opening.



Sealing valve in open position. The diffuser is flush with the oil drum opening.

Filling Line - Slide Valve

Top view of the slide valve.



Bottom view of the slide valve.



The screw in the U-panel serves as the front stop for the slide valve. In closed position, wooden rods of the slide valve are flush with the end of the supporting structure.



Detailed view of screw as front stop of slide valve.